

★ WEEK 3 · PART 1 of 3

AR, MA, and Wold

The Building Blocks of Time-Series Memory

Lag operator · $AR(p)$ · $MA(q)$ · stationarity & invertibility · ACF/PACF

Once a series is stationary (Week 2), how do we describe its memory? Part 1 introduces the two engines — autoregression and moving averages — and the fingerprints that tell them apart.

1

PART 1 · AR & MA

Two Engines of Memory

Autoregression, moving averages, the stationarity and invertibility conditions, Wold's theorem, and the ACF/PACF signatures that identify each.

The Friendship That Wrote the Rulebook — Part 1

How two statisticians gave time series a common language

George Box and Gwilym Jenkins turned a scattered collection of tricks into one disciplined method — a shared grammar for describing and forecasting any stationary series.

The two people

- George Box: a chemist-turned-statistician, famously practical.
- Gwilym Jenkins: a systems and control engineer.
- Together they bridged theory and real forecasting.

The problem they faced

- Practitioners had many ad-hoc models but no common language.
- Choosing among them was an art, not a procedure.
- They wanted a repeatable, teachable method.

The Friendship That Wrote the Rulebook — Part 2

The 1970 book, and the lesson for us

Their 1970 book gave us the ARMA family and the Box-Jenkins method — and Box's enduring reminder: "All models are wrong, but some are useful."

What they delivered

- A unified AR-MA-ARMA framework for stationary series.
- A disciplined loop: identify, estimate, diagnose, repeat.
- It remains the backbone of classical time-series analysis.

The lesson

- A model is a useful approximation, never the truth.
- Judge it by its residuals and its forecasts, not its elegance.
- This humility is the heart of the whole method.

The Big Idea: Describe a Series by Its Memory

Three ways the recent past can shape today

1

Today can echo its own past values

If high values tend to follow high values, the series remembers its own level. That memory is autoregression (AR).

2

Today can echo recent shocks

If a surprise today lingers for a few periods before fading, the series remembers shocks. That memory is a moving average (MA).

3

A series can do both at once

Combine the two and you get ARMA — a compact grammar flexible enough to describe almost any stationary series.

ARMA in one line: explain today using the recent past — its own values and its recent shocks.

White Noise: The Raw Material

Every ARMA model is white noise, filtered

White noise is a stream of independent, zero-mean shocks with no memory. AR and MA models are simply different ways of filtering this raw randomness into structure.

What it is

- Mean zero, constant variance, zero autocorrelation.
- The purest stationary process — no structure at all.
- Denoted ε_t ; the innovation or 'shock' each period.

Why it anchors everything

- AR and MA both take white noise as their input.
- A good model leaves white-noise residuals behind.
- So 'are the residuals white noise?' is the final test.

The Lag Operator

A compact language for the whole ARMA family

$$L X_t = X_{t-1}, \quad \phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$$

What it does

- L shifts a series back one step: $L X_t = X_{t-1}$.
- Powers stack: $L^2 X_t = X_{t-2}$, and so on.
- It turns lag structure into simple polynomials in L .

Why we bother

- AR and MA become polynomials $\phi(L)$ and $\theta(L)$.
- Stationarity and invertibility become root conditions.
- It makes the algebra of ARMA clean and mechanical.

AR(1): The Simplest Memory

Today is a fraction of yesterday, plus a shock

$$X_t = c + \phi X_{t-1} + \varepsilon_t \quad (\text{AR}(1))$$

How to read it

- ϕ is the persistence: how much of yesterday carries over.
- c sets the long-run mean; ε_t is the fresh shock.
- ϕ near 1 means long memory; ϕ near 0 means almost none.

The familiar face

- You met AR(1) in Week 2 as the unit-root borderline.
- Here $\phi < 1$ keeps it stationary and mean-reverting.
- It is the atom from which longer memory is built.

What AR(1) Actually Looks Like

Mean, variance, and autocorrelation in closed form

$$\mathbb{E}[X_t] = \frac{c}{1-\phi}, \quad \text{Var}(X_t) = \frac{\sigma^2}{1-\phi^2}, \quad \rho(k) = \phi^k$$

The three formulas

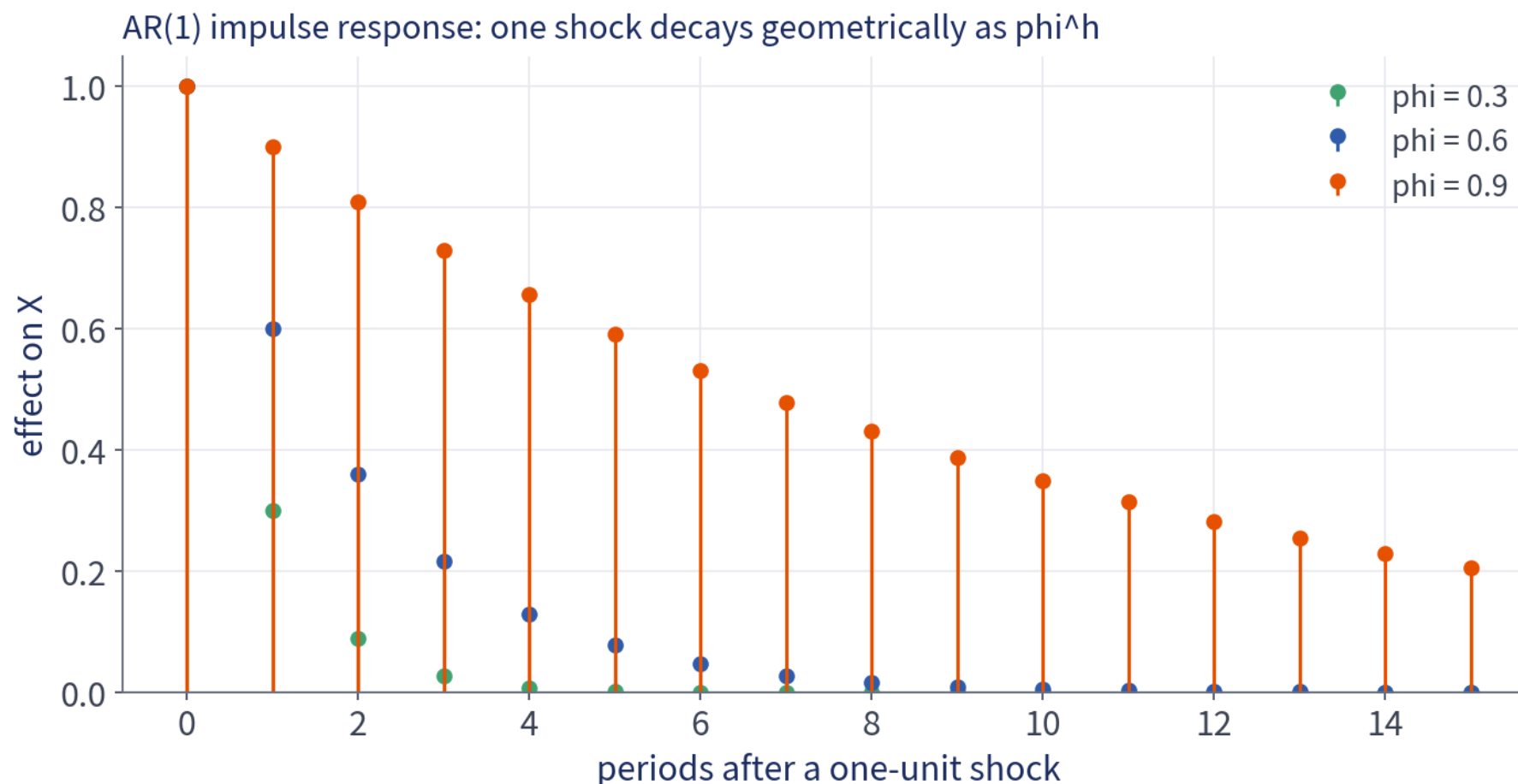
- Mean $c/(1-\phi)$: the level it hovers around.
- Variance grows as ϕ approaches 1 — more persistence, more spread.
- ACF $\rho(k) = \phi^k$: correlation decays geometrically with lag.

The key takeaway

- An AR(1)'s ACF tails off smoothly — it never cuts to zero.
- That smooth decay is the AR fingerprint.
- The next chart shows the same decay as a shock's echo.

How a Single Shock Propagates in an AR(1)

One unit shock today leaves an echo that decays geometrically as ϕ^h — larger ϕ means a longer-lived, slower-fading memory.



AR(p): Longer Memory

Let the last p values all have a say

$$X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t \iff \phi(L)X_t = c + \varepsilon_t$$

The generalization

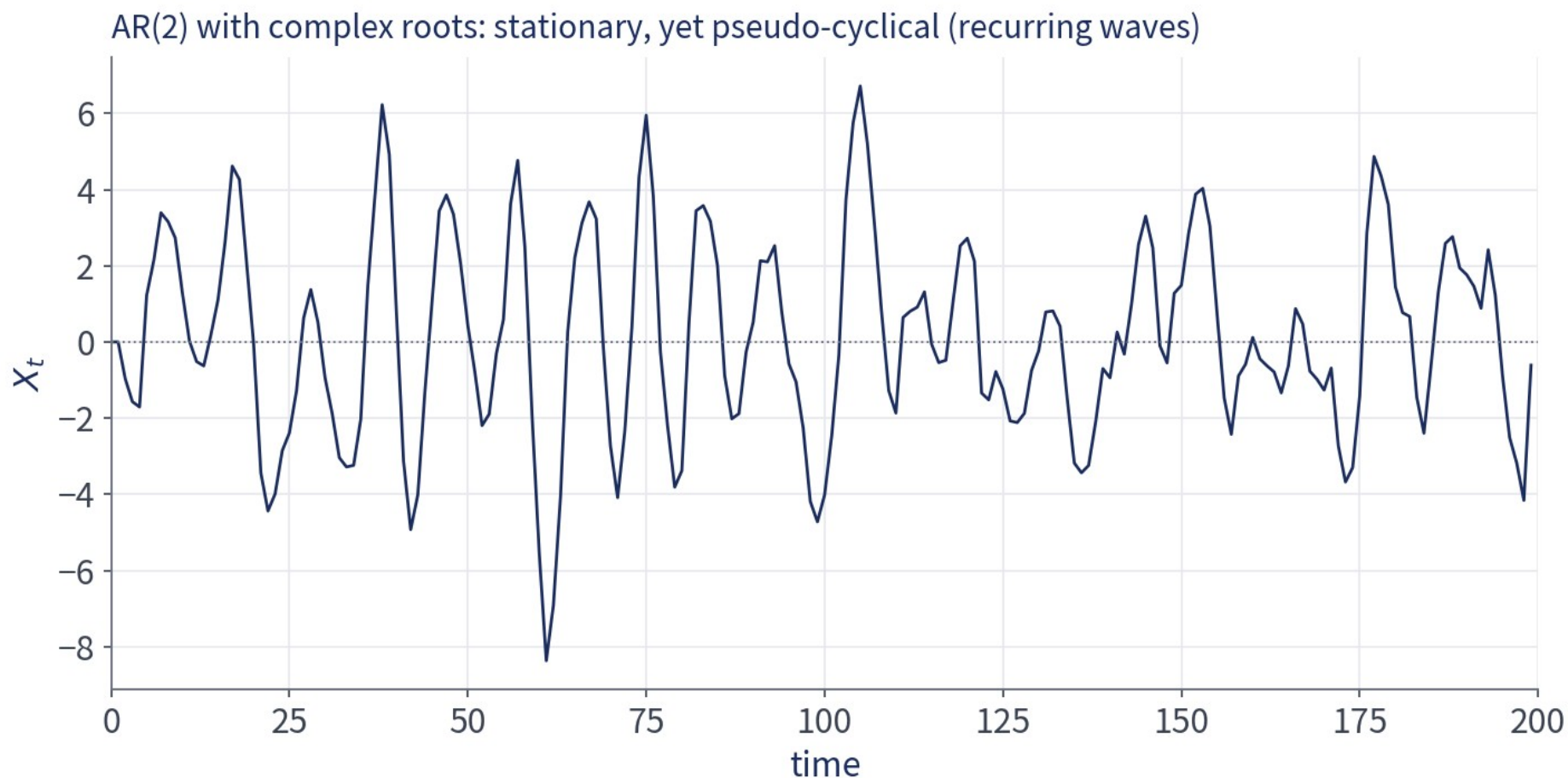
- Regress today on its last p values, each with its own weight.
- In lag-operator form it is simply $\phi(L)X_t = c + \varepsilon_t$.
- p is the order — how far back the direct memory reaches.

What p buys you

- Richer dynamics: overshooting, cycles, slow decay.
- AR(2) can even oscillate (next slide).
- But higher p means more parameters to estimate.

AR(2) Can Be Pseudo-Cyclical

When an AR(2)'s roots are complex, the series is still stationary yet rolls in recurring waves — memory that oscillates rather than simply decays.



When Is an AR Model Stationary?

The roots must lie outside the unit circle

stationary $\iff$ all roots of $\phi(z) = 0$ lie OUTSIDE the unit circle ($|z| > 1$)

The rule

- Write the AR polynomial $\phi(z)$ and find its roots.
- All roots must lie strictly outside the unit circle.
- For AR(1) this reduces exactly to $|\phi| < 1$.

Why it matters

- A root ON the circle is a unit root — non-stationary (Week 2).
- A root INSIDE means an explosive, useless process.
- Every stats package checks this for you automatically.

MA(1): A Finite Echo of Shocks

Surprises matter, but only for a while

$$X_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}, \quad \rho(1) = \frac{\theta}{1+\theta^2}, \quad \rho(k) = 0 \quad (k \geq 2)$$

How to read it

- Today is this period's shock plus a fraction of last period's.
- The influence of any shock lasts exactly one lag, then stops.
- θ sets how strongly the previous surprise carries over.

The signature

- Its ACF is non-zero at lag 1, then exactly zero after.
- That sharp cut-off is the MA fingerprint.
- Contrast with AR's smooth, never-ending decay.

MA(q): Echoes of the Last q Shocks

A longer but still finite memory

$$X_t = \mu + \theta(L) \varepsilon_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

The generalization

- Today is a weighted sum of the last q shocks, plus today's.
- In lag form: $X_t = \mu + \theta(L) \varepsilon_t$.
- q is the memory length — how many lags a shock survives.

The clean property

- An MA(q)'s ACF cuts off sharply after lag q.
- So the last non-zero ACF lag reveals q directly.
- This is what makes MA models easy to identify.

Invertibility: The MA Counterpart

So the model has a unique, sensible form

invertible $\iff$ all roots of $\theta(z) = 0$ lie OUTSIDE the unit circle

What it means

- An invertible MA can be rewritten as an infinite AR.
- Its roots, like AR's, must lie outside the unit circle.
- This lets us recover the shocks from the observed data.

Why we require it

- Different θ values can give the SAME autocorrelations.
- Invertibility picks the one, well-behaved representation.
- Without it, estimation and forecasting are ambiguous.

Why MA Is So Fundamental: Wold's Theorem

Every stationary series is an MA at heart

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \psi_0 = 1 \quad (\text{Wold representation})$$

What it says

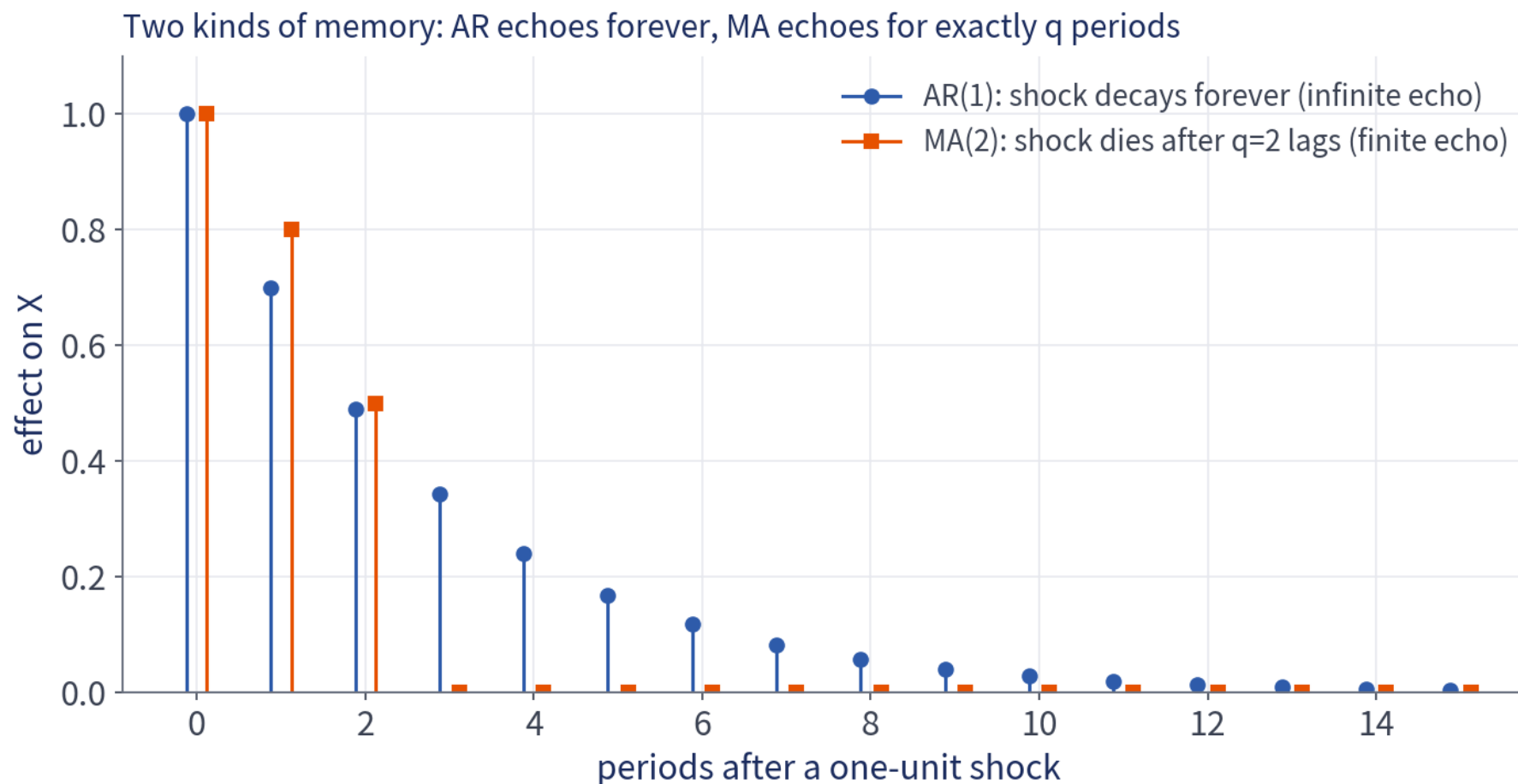
- Any (weakly) stationary series equals an $\text{MA}(\infty)$ of white noise.
- So moving averages are not just one option — they are universal.
- AR and ARMA are just compact ways to write that $\text{MA}(\infty)$.

Why it is reassuring

- It guarantees the ARMA toolbox is broad enough.
- It explains the $\text{AR} \leftrightarrow \text{MA}$ duality we see next.
- It is the theoretical license for the whole method.

AR vs MA: Two Kinds of Memory

Follow a single shock: in an AR it echoes forever, decaying geometrically; in an MA(q) it lives for exactly q periods and then vanishes.



AR and MA Are Two Sides of a Coin

A stationary AR is an infinite MA, and vice versa

$$\underbrace{X_t = \phi X_{t-1} + \varepsilon_t}_{\text{AR}(1)} = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j} = \underbrace{\text{MA}(\infty)}_{\text{same process}}$$

The equivalence

- Unrolling an AR(1) turns it into an MA(∞) of past shocks.
- Likewise an invertible MA(1) becomes an AR(∞).
- The two descriptions are the same process, written differently.

Why it is useful

- It lets you pick whichever form is more parsimonious.
- It explains why the ACF and PACF are mirror images.
- It is Wold's theorem made concrete.

The ACF and the PACF

Two views of a series' memory

The ACF measures the TOTAL correlation at each lag; the PACF measures the DIRECT correlation, stripping out what nearer lags already explain. Together they identify a model.

ACF — total correlation

- How correlated is X_t with X_{t-k} , in full?
- For an $MA(q)$ it cuts off sharply after lag q .
- For an AR it tails off smoothly.

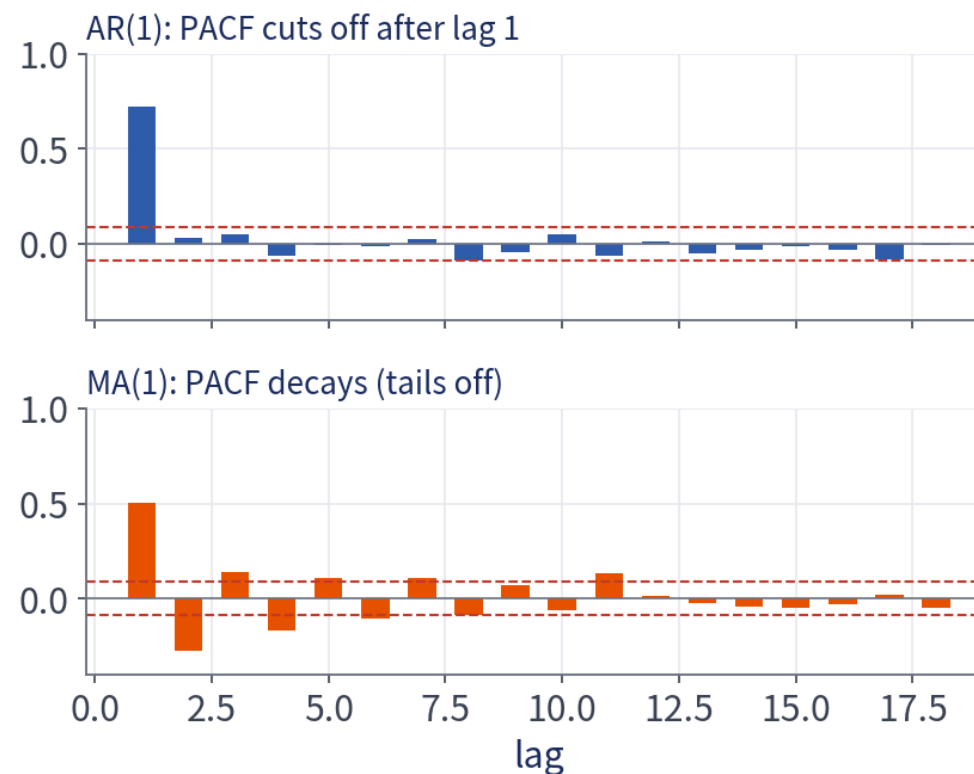
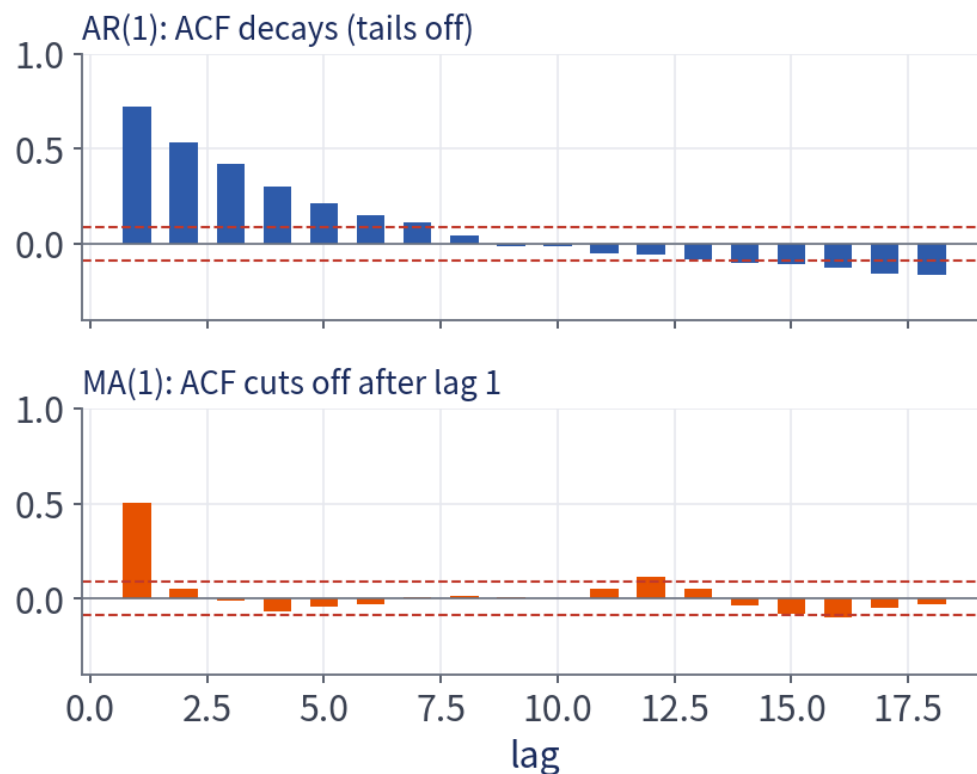
PACF — direct correlation

- The link at lag k after removing lags $1 \dots k-1$.
- For an $AR(p)$ it cuts off sharply after lag p .
- For an MA it tails off smoothly.

ACF / PACF Signatures at a Glance

The single most useful picture in classical time series: whichever plot CUTS OFF tells you the model — ACF cut-off $\rightarrow$ MA(q); PACF cut-off $\rightarrow$ AR(p).

The identification fingerprint: which one cuts off tells you AR vs MA



AR vs MA: Two Engines of Memory

Both turn white-noise shocks into structure, but they remember in opposite ways — and leave opposite fingerprints on the ACF and PACF.

AR vs MA: Two Engines of Memory

Both turn white-noise shocks into structure - but they remember in opposite ways.

AR - Autoregressive (feedback)

Today depends on its own PAST VALUES.

$$X_t = c + \phi X_{t-1} + \dots + e_t$$

One shock feeds back through the level, so its echo decays but never fully dies.

Fingerprint

- ACF: decays (tails off)
- PACF: cuts off after lag p

memory = infinite, geometric

MA - Moving Average (echo)

Today depends on recent SHOCKS.

$$X_t = \mu + e_t + \theta e_{t-1} + \dots$$

A shock echoes for exactly q periods, then vanishes completely.

Fingerprint

- ACF: cuts off after lag q
- PACF: decays (tails off)

memory = finite, exactly q lags

Simulate and Identify

Build the AR/MA fingerprints in your own hands

GOAL Simulate AR and MA processes, then learn to read their ACF/PACF fingerprints before touching real data.

[INPUT] Prompt · Claude Code

Write and run `arma_signatures.py` that:

1. simulates four series of length 800 -
AR(1) $\phi=0.7$, AR(2) with complex roots,
MA(1) $\theta=0.8$, and MA(2);
2. for each, plots the series, its ACF, and
its PACF (with 95% bands) in one row;
3. prints, for each, which plot cuts off and
at what lag - and its implied (p, q).

Then hide the labels and quiz me: from the
ACF/PACF alone, can I name each model?

How to work with it

- AR $\rightarrow$ PACF cuts off; MA $\rightarrow$ ACF cuts off.
- Read the cut-off lag as p or q.
- Re-run with new seeds — do the signatures hold?
- Keep it; Part 2 uses these to identify real data.

Deliverable: `arma_signatures.py` + a one-row ACF/PACF panel per process.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

A series' ACF cuts off after lag 2 but its PACF tails off. AR or MA, what order?

MA(2)

Q2

Why can every stationary series be written as a moving average?

Wold's theorem

Q3

An AR(1) has $\phi = 1.2$. What is wrong with it?

Explosive (root inside)

All three come straight from the AR/MA fingerprints and the root conditions.

Three Misconceptions to Avoid

Where the fingerprints get misread

Common myths

- "A slowly-decaying ACF means a high MA order."
- "AR and MA are fundamentally different processes."
- "Bigger p and q always fit better."

The reality

- Slow ACF decay points to AR, not MA — read the PACF to confirm.
- By duality they are two descriptions of one process (Wold).
- More parameters overfit; parsimony wins (Part 3's AIC/BIC).

Three Things to Carry Forward

Part 1 in one screen

1

Two engines: AR remembers levels, MA remembers shocks

AR feeds back through past values; MA echoes recent innovations. Stationarity (AR) and invertibility (MA) both come down to roots outside the unit circle.

2

Wold ties it together

Every stationary series is an $MA(\infty)$, and $AR \leftrightarrow MA$ duality means the two forms describe the same process — pick the compact one.

3

The ACF and PACF are your fingerprints

ACF cut-off $\rightarrow MA(q)$; PACF cut-off $\rightarrow AR(p)$. That single reading is the key to identification in Part 2.

Next — Part 2: combining both into ARMA, and the Box-Jenkins method for identifying p and q .

★ WEEK 3 · PART 1 · PRACTICE

AR, MA, and Wold, Worked by Hand

Read a Series' Memory in Closed Form, Then Tell AR From MA

AR(1) properties, MA(1) cutoff, and the AR-MA duality

Part 1's building blocks, with the arithmetic shown. We compute an AR(1)'s mean, variance, and autocorrelation in closed form, then an MA(1)'s variance and its sharp ACF cutoff - and see how a stationary AR is secretly an infinite MA, the heart of Wold's theorem.

1P

PART 1 · APPLICATIONS

Memory, in Closed Form

Case A: an AR(1)'s mean, variance, and geometrically decaying autocorrelation, plus its stationarity condition and memory half-life. Case B: an MA(1)'s variance and sharp ACF cutoff, and the AR-MA duality that makes every stationary series an MA at heart. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

A series is white noise, filtered by memory

AR carries yesterday's value forward; MA carries yesterday's shock. Both build structure from pure noise.

2

AR autocorrelation decays; MA cuts off

An AR(1)'s ACF fades geometrically and never reaches zero; an MA(1)'s is a single spike, then exactly zero.

3

AR and MA are two sides of a coin

A stationary AR is an infinite MA and vice versa - Wold's theorem: every stationary series is an MA at heart.

Everything here is a calculator exercise: one AR in closed form, one MA cutoff, one duality.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from an AR or MA coefficient.
- Compute the closed-form properties.
- Then read the memory.

What you should watch

- How the AR autocorrelation decays.
- How the MA autocorrelation cuts off.
- How an AR becomes an infinite MA.

The AR(1) Memory

Today is a fraction of yesterday

A mean-reverting return follows an AR(1): today is a fraction of yesterday plus a fresh shock. That single coefficient fixes everything - the long-run mean, the variance, and how autocorrelation decays. We compute all three in closed form, confirm stationarity, and read off the memory's half-life

The model

- Coefficient ϕ of 0.6.
- Constant of 0.4, shock variance 1.0.
- A persistent, mean-reverting series.

What we'll do by hand

- Compute the mean and variance.
- Compute the autocorrelation at each lag.
- Find the half-life of the memory.

The AR(1) Model

A fraction of yesterday, plus a shock

$$y_t = c + \phi y_{t-1} + \varepsilon_t \quad (\text{today is a fraction } \phi \text{ of yesterday, plus a fresh shock})$$

The pieces

- Phi scales yesterday's value into today.
- The constant sets the long-run level.
- The shock is fresh white noise.

The intuition

- A high phi means a long memory.
- A low phi means the past fades fast.
- Below one keeps it stationary.

Mean and Variance

In closed form

$$\mu = \frac{c}{1-\phi} = \frac{0.4}{0.4} = \mathbf{1.0}, \quad \gamma_0 = \frac{\sigma^2}{1-\phi^2} = \frac{1.0}{0.64} = \mathbf{1.5625}$$

The mean

- The constant over one minus phi.
- 0.4 over 0.4 equals 1.0.
- That's the level it reverts to.

The variance

- The shock variance over one minus phi-squared.
- 1.0 over 0.64 equals 1.5625.
- Larger than the shock, because of persistence.

Autocorrelation and Stationarity

A geometric decay

$$\rho_k = \phi^k : 0.6, 0.36, 0.216, \dots \text{ (geometric decay); stationary } \iff |\phi| < 1$$

The decay

- The autocorrelation at lag k is ϕ to the k .
- 0.6, 0.36, 0.216, and so on.
- It fades but never quite reaches zero.

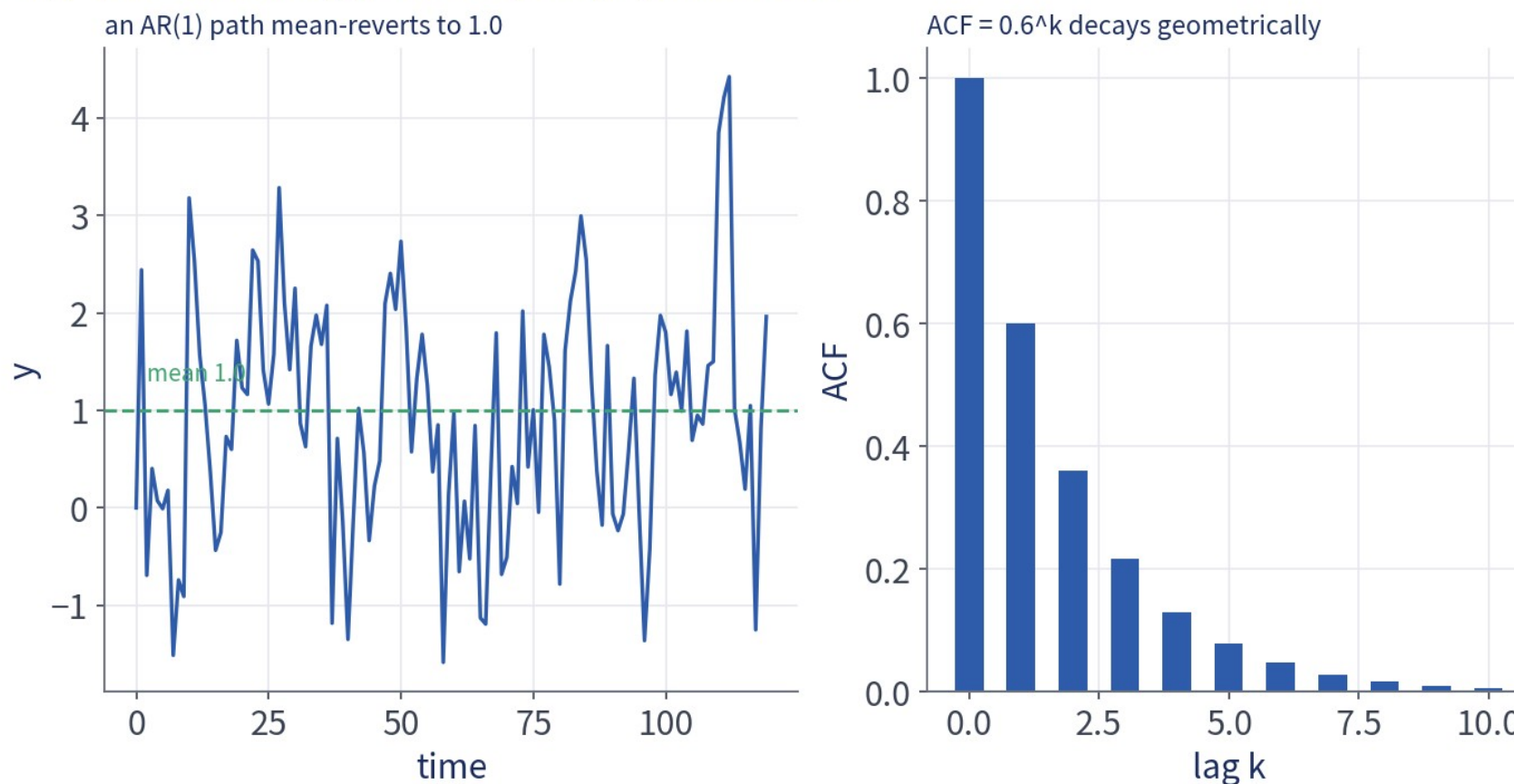
Half-life and stability

- The correlation halves in about 1.4 lags.
- Stationary because ϕ is below one.
- At one or above, it would never settle.

Persistence and Its Decay

On the left, an AR(1) path wanders but keeps returning to its mean of 1.0 - the pull of mean reversion. On the right, its autocorrelation function: a smooth geometric decay of 0.6 to the k, fading toward zero without ever reaching it. That never-quite-zero tail is the signature of an AR process.

Case A: AR(1) - persistent memory, geometrically decaying autocorrelation



Reading the Memory

What ϕ tells a trader

What ϕ encodes

- The strength of mean reversion.
- How fast a shock's effect fades.
- The half-life of predictability.

Why it matters

- A high ϕ means trends persist - momentum.
- A low or negative ϕ means quick reversal.
- The ACF's decay rate is the trading horizon.

MA(1) and the Duality

A finite echo, and Wold's coin

An MA(1) carries yesterday's shock, not yesterday's value - a finite echo. Its autocorrelation is a single spike, then exactly zero, a sharp contrast with the AR's slow decay. We compute the MA's variance and that cutoff, check invertibility, then reveal the duality: a stationary AR(1) is secretly an infinite MA - the essence of Wold's

or

The MA(1)

- Today's value is a shock plus part of yesterday's.
- Theta of 0.8, shock variance 1.0.
- A memory just one period long.

What we'll do by hand

- Compute its variance and lag-1 autocorrelation.
- Confirm the cutoff and invertibility.
- Write the AR(1) as an infinite MA.

The MA(1) Model

A finite echo of one shock

$$\gamma_0 = \sigma^2(1 + \theta^2) = \mathbf{1.64}, \quad \rho_1 = \frac{\theta}{1+\theta^2} = \frac{0.8}{1.64} = \mathbf{0.49}, \quad \rho_{k \geq 2} = \mathbf{0}$$

The variance

- The shock variance times one plus theta-squared.
- One times 1.64, so 1.64.
- Inflated by the echoed shock.

The cutoff

- Lag-1 autocorrelation is 0.49.
- Every lag beyond is exactly zero.
- A one-period memory, sharply bounded.

The Sharp Cutoff

And invertibility

lag k	1	2	3	4	5
MA(1) ACF	0.49	0	0	0	0
AR(1) ACF	0.60	0.36	0.22	0.13	0.08

The MA(1) autocorrelation is 0.49 at lag one and exactly zero everywhere after - a sharp cutoff. The AR(1) beneath it decays but never dies. That contrast is what identifies one from the other. The MA is invertible because θ is below one.

AR Equals Infinite MA

Wold's theorem, concretely

$$\text{AR}(1) \ \phi=0.6 \ = \ \text{MA}(\infty) : \ y_t = \sum_{j \geq 0} \phi^j \varepsilon_{t-j} = \varepsilon_t + 0.6\varepsilon_{t-1} + 0.36\varepsilon_{t-2} + \dots$$

The expansion

- Substitute the AR(1) into itself repeatedly.
- Each shock enters with weight ϕ to its age.
- 1, 0.6, 0.36, 0.216, fading away.

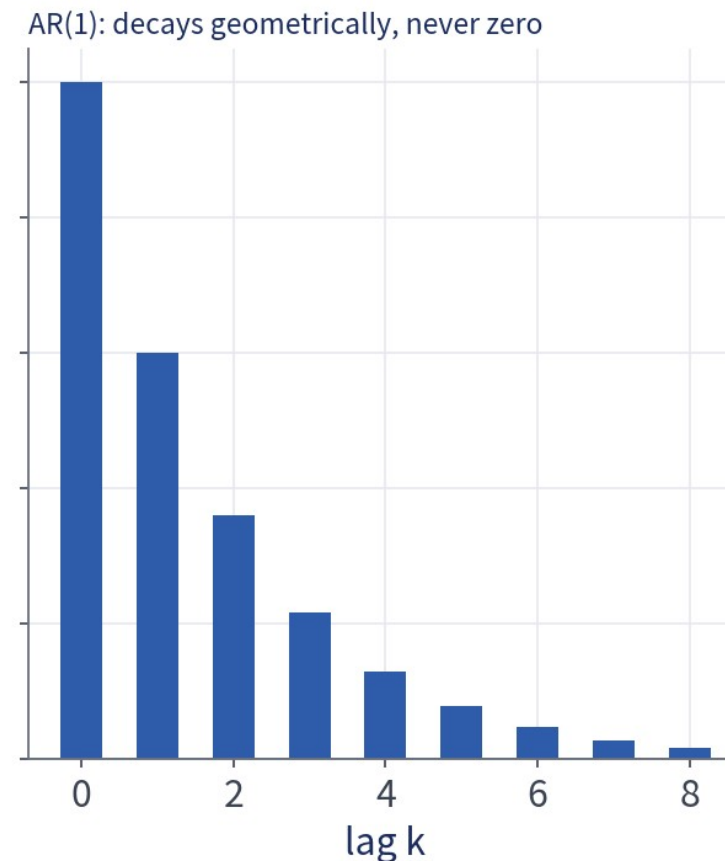
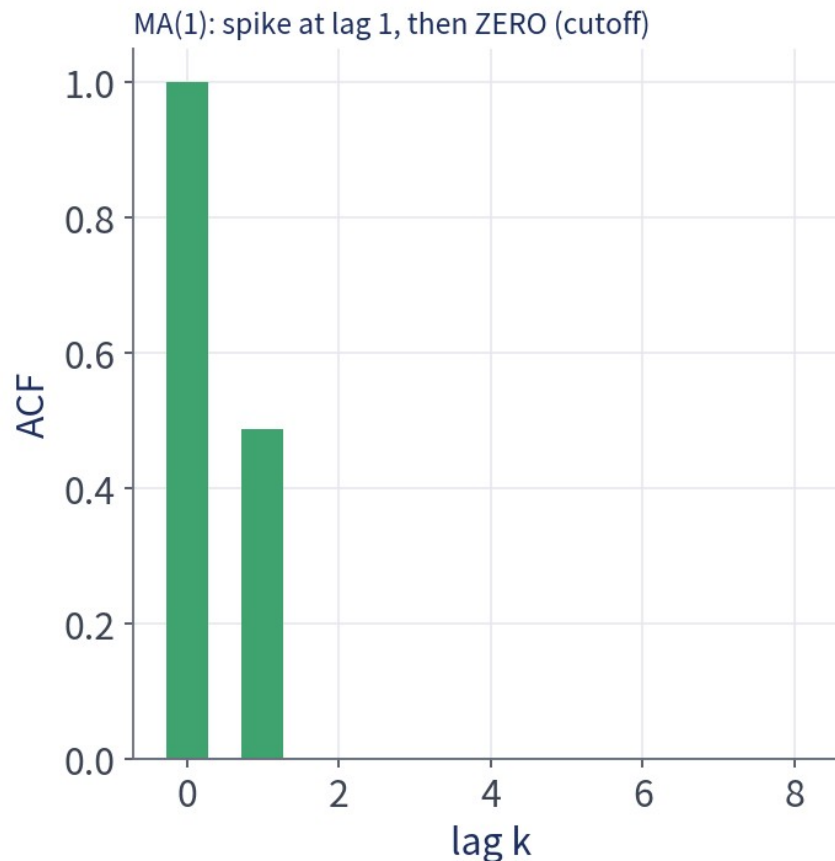
Why it matters

- A stationary AR is an infinite MA.
- Wold: every stationary series is an MA at heart.
- AR and MA are two views of one thing.

Cutoff Versus Decay

Side by side, the two signatures. The MA(1) autocorrelation (left) spikes at lag one and drops to exactly zero - a clean cutoff. The AR(1) (right) decays geometrically, tapering but never vanishing. Reading which pattern a series shows is how you tell an MA from an AR - the whole basis of identification.

Case B: cutoff vs decay - the signature that tells MA from AR



The Signature That Identifies

Cutoff, decay, and Wold

The identification rule

- A sharp ACF cutoff points to MA.
- A slow ACF decay points to AR.
- Next week turns this into a full method.

Why Wold matters

- Any stationary series has an MA form.
- So MA models are universal building blocks.
- AR is just a compact way to write an infinite MA.

The Two Tools

Memory, persistence, and correlation

AR and MA are the two atoms of linear time series. AR builds persistence from past values, giving an autocorrelation that decays; MA builds a finite echo from past shocks, giving one that cuts off. Their closed-form properties - mean, variance, autocorrelation - follow from a single coefficient each. And the duality binds them:

ie

AR(1) (Case A)

- Mean 1.0, variance 1.5625.
- ACF 0.6 to the k - geometric decay.
- Half-life about 1.4 lags.

MA(1) and duality (Case B)

- Variance 1.64, ACF 0.49 then zero.
- AR(1) equals an infinite MA.
- Wold: every stationary series is MA.

Reproduce the Properties and the Duality

Direct Claude Code to check your arithmetic

GOAL Compute AR(1) and MA(1) properties, then expand the AR as an infinite MA.

[INPUT] Prompt · Claude Code

```
Write and run part1_practice.py that:  
CASE A - AR(1), phi=0.6, c=0.4, sigma^2=1:  
  * print the mean, variance, and ACF for lags 1-5;  
  * print the memory half-life.  
CASE B - MA(1), theta=0.8:  
  * print the variance and the lag-1 ACF;  
  * confirm ACF is zero for lag >= 2;  
  * expand AR(1) as an MA and print psi_0..psi_4.  
Then plot the AR path with its ACF, and the MA-vs-AR  
ACF comparison.
```

Check against the slides

- AR mean 1.0, variance 1.5625.
- AR ACF 0.6, 0.36, 0.216, ...
- MA variance 1.64, ACF 0.49.
- $\psi_j = 0.6^j$.

Deliverable: part1_practice.py whose printed numbers match the AR and MA properties.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How does an AR(1)'s autocorrelation behave?

**It decays
geometrically as ϕ
to the k**

Q2 What's distinctive about an MA(1)'s ACF?

**A single spike at lag
one, then exactly
zero**

Q3 What does Wold's theorem say?

**Every stationary
series is an infinite
MA at heart**

All three come from the AR closed form, the MA cutoff, and the AR-MA duality.

Three Pitfalls to Avoid

Where AR and MA trip people

Common mistakes

- "An AR autocorrelation eventually hits zero."
- "MA and AR ACFs look the same."
- "Any ϕ gives a stationary AR."

The reality

- It decays toward zero but never exactly reaches it.
- MA cuts off sharply; AR decays - that's the tell.
- Stationarity needs ϕ below one in magnitude.

Two Calculations to Remember

Part 1 applications in one screen

1

AR(1) properties in closed form

With ϕ of 0.6, the mean was 1.0, the variance 1.5625, and the autocorrelation 0.6 to the k - 0.6, 0.36, 0.216 - a geometric decay with a half-life near 1.4 lags. One coefficient fixes the entire memory.

2

MA(1) cuts off; AR is an infinite MA

The MA(1) had variance 1.64 and a lag-1 autocorrelation of 0.49, then exactly zero - a sharp cutoff. And the AR(1) expanded into an infinite MA with weights 1, 0.6, 0.36, illustrating Wold: every stationary series is an MA at heart.

3

Cutoff versus decay is the tell

A sharp ACF cutoff signals MA; a slow decay signals AR. That single contrast, plus the AR-MA duality, is the foundation the next part builds identification on.

Next — Part 2 practice: turning the ACF and PACF signatures into a full ARMA identification method.

★ WEEK 3 · PART 2 of 3

ARMA and Identification

Combining the Engines, and the Box-Jenkins Method

ARMA(p,q) · the ACF/PACF cheat sheet · the identify–estimate–diagnose loop

Part 1 gave us AR and MA separately. Part 2 fuses them into ARMA and turns identification into a disciplined procedure — the Box-Jenkins method that made time-series modeling teachable.

2

PART 2 · IDENTIFICATION

Choosing p and q , With Discipline

ARMA in one framework, the ACF/PACF signatures that reveal the order, and the iterative Box-Jenkins loop from raw data to a validated model.

From Two Engines to One Method

Part 1 gave the parts; Part 2 assembles them

1

We can now read AR and MA fingerprints

PACF cut-off means AR; ACF cut-off means MA. That single reading is the foundation everything here builds on.

2

Real series usually need both

Combining feedback (AR) and shock echoes (MA) yields ARMA — often a far more compact fit than either alone.

3

Identification becomes a procedure

Box and Jenkins turned 'which model?' into a repeatable loop: identify, estimate, diagnose, and repeat until the residuals are clean.

Part 2 in one line: combine AR and MA, then use the ACF/PACF to choose p and q .

ARMA(p,q): Combining Both

Feedback and shock echoes in one model

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}$$

The two halves

- The AR part: today depends on its own past p values.
- The MA part: today also echoes the last q shocks.
- Together they describe both persistence and short-lived surprises.

Reading (p, q)

- p is the autoregressive order; q is the moving-average order.
- ARMA(1,0) is AR(1); ARMA(0,1) is MA(1).
- Most well-behaved series need only small p and q .

ARMA in Lag-Operator Form

One compact line captures the whole family

$$\phi(L) X_t = c + \theta(L) \varepsilon_t \quad (\text{ARMA in lag-operator form})$$

Why this form

- $\phi(L)$ is the AR polynomial; $\theta(L)$ is the MA polynomial.
- The whole model is one polynomial equation in L .
- Stationarity and invertibility are just root conditions on each.

What it unifies

- AR is $\theta(L) = 1$; MA is $\phi(L) = 1$; ARMA has both non-trivial.
- It makes the algebra of forecasting mechanical.
- Every stats package works in exactly this representation.

Why Combine? Parsimony

A small ARMA can replace a large pure model

A long-memory series might need an AR(10) — or an ARMA(1,1) with just two parameters. Combining the engines often buys the same fit with far fewer coefficients.

The trade you want

- Fewer parameters mean less to estimate and less to overfit.
- A tight ARMA generalizes better than a bloated AR or MA.
- This is the whole reason the ARMA class exists.

The cost

- ARMA is harder to identify — both plots tail off.
- So identification takes more judgment than pure AR or MA.
- The cheat sheet and the loop are how we manage that.

Choosing p and q

Let the data's autocorrelations guide you

Identification means reading the sample ACF and PACF to propose a candidate (p, q) . It is the first — and most skill-dependent — step of the whole method.

What we look for

- Where does the ACF cut off, tail off, or die slowly?
- Where does the PACF cut off or tail off?
- The pattern points to AR, MA, or ARMA — and its order.

Why it is hard

- Sample plots are noisy — no cut-off is perfectly sharp.
- ARMA makes both plots tail off, hiding the order.
- So we propose, fit, and check — never guess once.

The Identification Cheat Sheet

Memorize this — you will use it constantly

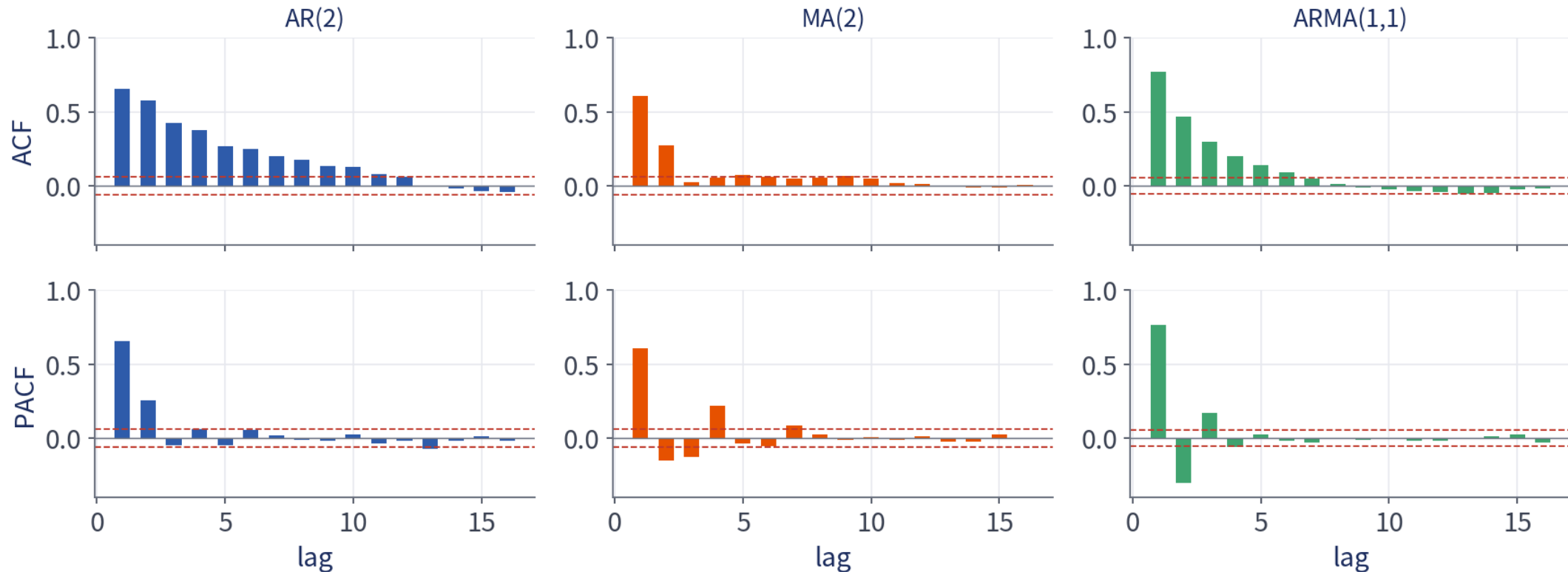
Model	ACF (total)	PACF (direct)
AR(p)	tails off (geometric / damped)	cuts off sharply after lag p
MA(q)	cuts off sharply after lag q	tails off (geometric / damped)
ARMA(p,q)	tails off	tails off

The rule: the plot that CUTS OFF names the order. If BOTH tail off, you are looking at an ARMA.

The Cheat Sheet, Visualized

AR(2) cuts its PACF; MA(2) cuts its ACF; ARMA(1,1) tails off in BOTH — the tell-tale that neither pure model will do.

Signature gallery: AR cuts PACF, MA cuts ACF, ARMA tails off in BOTH



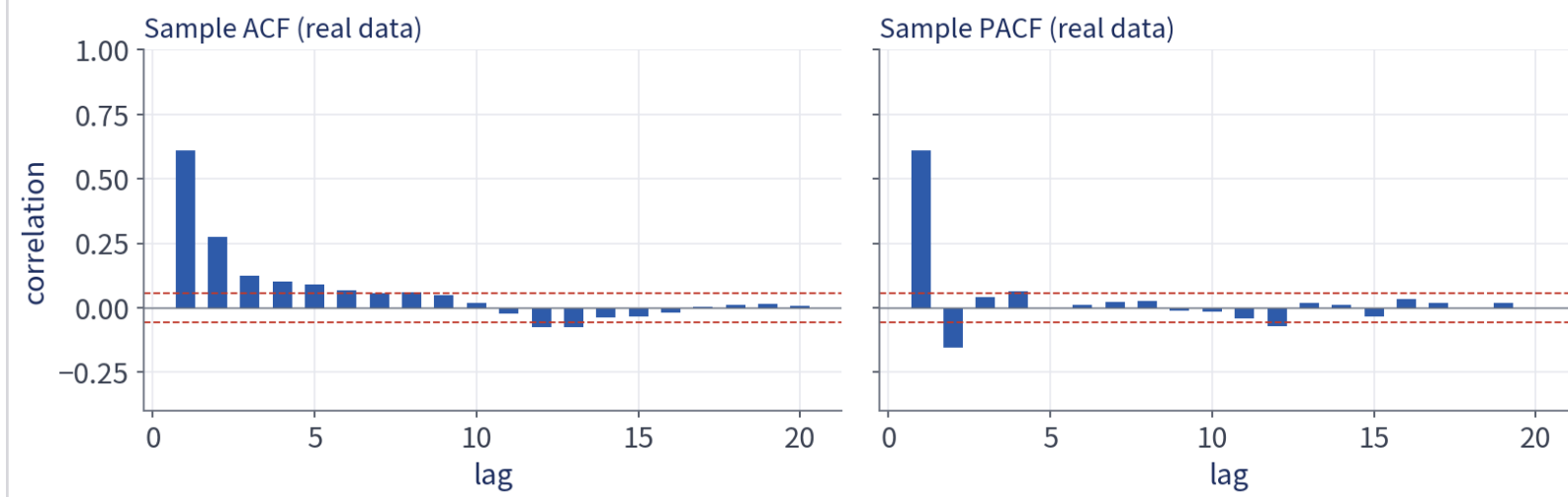
Real Data Is Never So Tidy

On real series, sampling noise blurs the cut-offs — identification becomes a judgment call, not a lookup.

What to expect

- No razor-sharp cut-off.
- Bars hover near the band.
- Several (p,q) look plausible.
- So propose a few.
- Then let diagnostics decide.

Real data is noisy: no clean cut-off — identification is a judgment call



The Box-Jenkins Methodology

A disciplined loop, not a one-shot recipe

Box-Jenkins replaces guesswork with a cycle: identify a candidate model, estimate it, check its residuals, and repeat until nothing systematic is left behind.

Why a loop

- The first candidate is rarely the final model.
- Diagnostics reveal what identification missed.
- Each pass refines p and q toward a clean fit.

The stopping rule

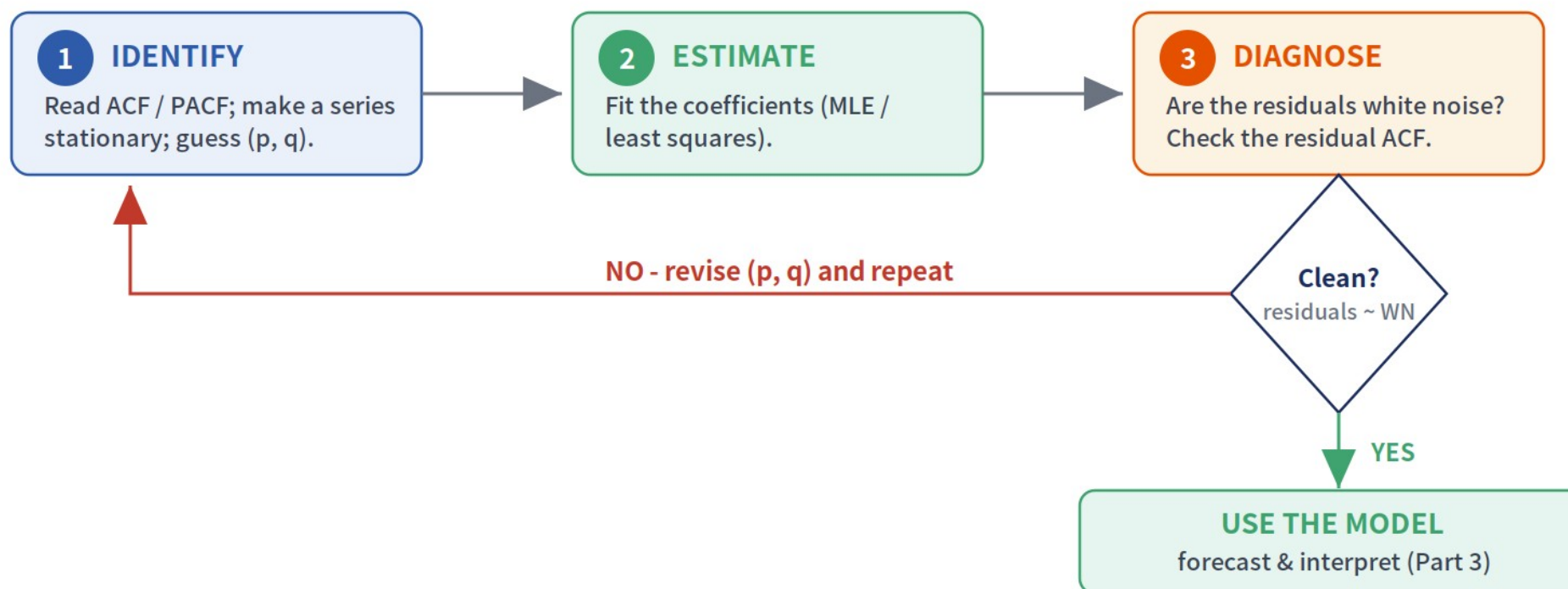
- Stop when the residuals look like white noise.
- At that point the model has captured all the structure.
- Then — and only then — do you forecast.

Box-Jenkins: The Iterative Loop

Identify, estimate, diagnose — and if the residuals aren't clean, revise (p, q) and go around again. Only white-noise residuals let you exit to forecasting.

The Box-Jenkins Loop

Not a one-shot recipe - a disciplined cycle you repeat until the residuals are clean.



Identification in Practice

Turning plots into a candidate model

The routine

- Make the series stationary first (Week 2): difference or detrend.
- Plot the sample ACF and PACF out to a couple dozen lags.
- Match the pattern to the cheat sheet and propose (p, q) .

Practical tips

- Keep p and q small to start — parsimony first.
- When both plots tail off, try a low-order ARMA like $(1,1)$.
- Note two or three candidates; you will compare them later.

Estimation in Brief

How the coefficients are actually fit

$$\hat{\theta} = \arg \min_{\theta} \sum_t \hat{\varepsilon}_t^2 \quad \text{or} \quad \arg \max_{\theta} \mathcal{L}(\theta)$$

The idea

- Choose coefficients that make the model's errors as small as possible.
- Equivalently, maximize the likelihood of the observed data.
- The software solves this numerically — you read the output.

What you get back

- Estimated ϕ 's and θ 's, each with a standard error.
- A log-likelihood you will reuse for AIC and BIC.
- Residuals to feed into the diagnostic step.

Estimation, Intuitively

What the optimizer is really doing

The fit hunts for the coefficients under which the model would have predicted the past best — the ones that leave the smallest, most structureless residuals.

Least squares view

- Every candidate model implies one-step-ahead predictions.
- Sum the squared prediction errors across the sample.
- The best coefficients minimize that sum.

Likelihood view

- Assume the shocks are (roughly) normal.
- Pick coefficients that make the data most probable.
- For Gaussian shocks the two views nearly coincide.

Diagnostic Checking

The gate you must pass before trusting a model

A good ARMA leaves residuals that are indistinguishable from white noise. If the residual ACF still shows structure, the model has missed something — go back and revise.

What to check

- Plot the residual ACF — no spikes should stand out.
- Run a portmanteau test (e.g. Ljung-Box) on the residuals.
- Look for remaining patterns, outliers, or changing variance.

How to respond

- A leftover spike at lag k suggests raising p or q .
- Changing variance points to a volatility model (later weeks).
- Iterate until the residuals are clean — that is the exit.

Parsimony: The Guiding Principle

The simplest model that works, wins

Between two models that fit equally well, always prefer the one with fewer parameters. Extra coefficients fit noise, not signal — and noise does not repeat out of sample.

Why simpler is safer

- Fewer parameters mean smaller estimation error.
- Simple models generalize; complex ones memorize the sample.
- This is the bias-variance trade-off in one sentence.

How we enforce it

- Start small and add order only when diagnostics demand it.
- Penalize complexity explicitly with AIC and BIC (next).
- Box's warning again: a useful model, not a perfect one.

Common ARMA Specifications

A short menu that covers most cases

Model	What it captures	Typical use
AR(1)	simple persistence / mean reversion	interest rates, spreads, ratios
MA(1)	a one-period shock echo	microstructure noise, revisions
ARMA(1,1)	persistence plus a short echo	many macro & financial series
AR(2)	cycles and overshooting	business-cycle-like dynamics

Start with one of these; escalate order only if the residuals still show structure.

Model Selection: AIC and BIC

Balancing fit against complexity

$$\text{AIC} = -2 \ln \hat{\mathcal{L}} + 2k, \quad \text{BIC} = -2 \ln \hat{\mathcal{L}} + k \ln T$$

How to read them

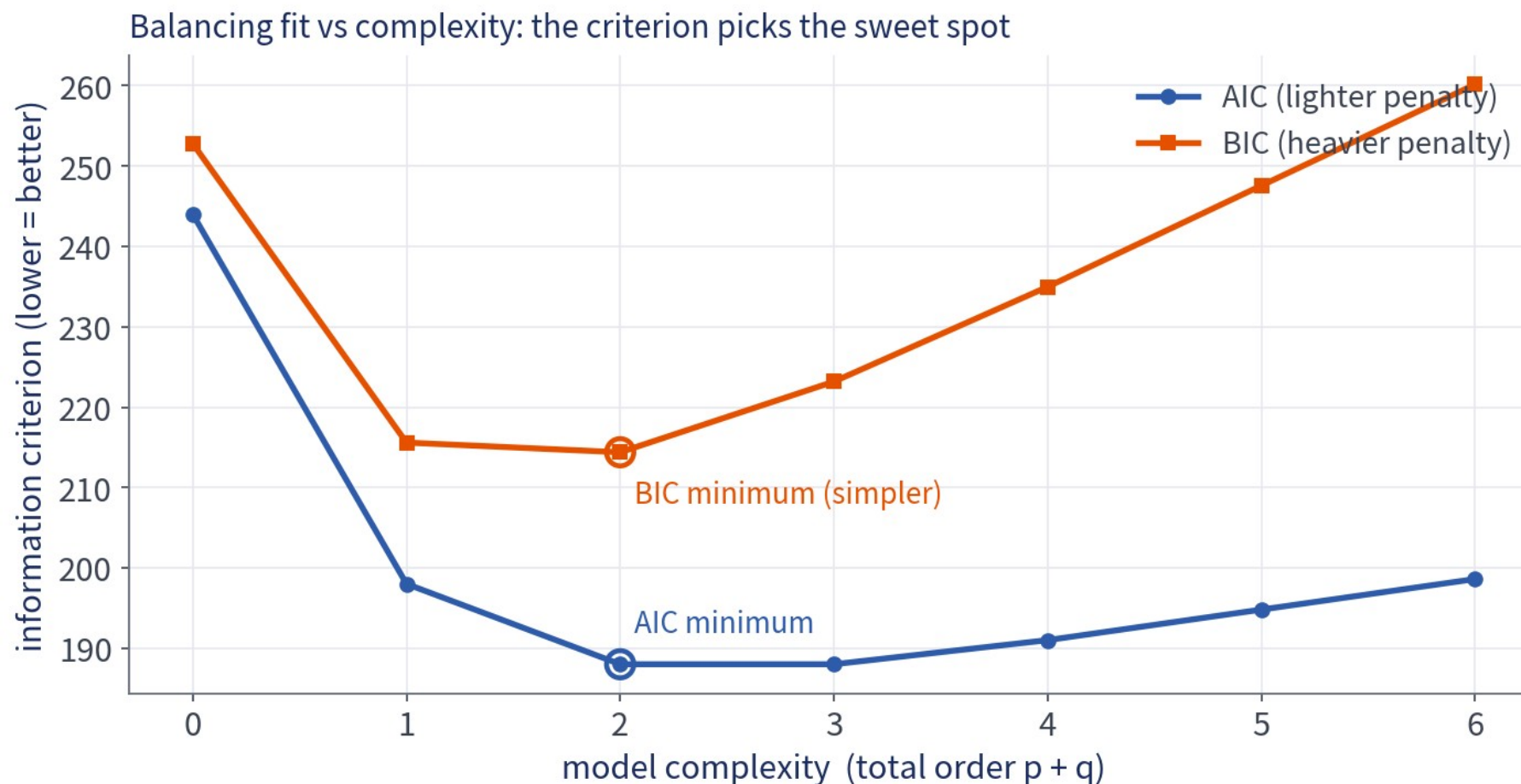
- Both reward fit (the likelihood) and penalize the parameter count k .
- Lower is better; compare candidates and pick the minimum.
- BIC's penalty ($k \cdot \ln T$) is heavier, so it favors simpler models.

How to use them

- Fit a small grid of (p, q) candidates.
- Rank by AIC or BIC and shortlist the best.
- Then confirm the winner with residual diagnostics.

Fit vs Complexity, Made Visible

As order grows, fit improves but the penalty rises — the criterion dips to a minimum. BIC's heavier penalty lands on a simpler model than AIC.



Overfitting: The Ever-Present Danger

When more fit means worse forecasts

Add enough parameters and any model fits the past perfectly — including its noise. That noise will not recur, so the over-fit model forecasts worse, not better.

The symptoms

- In-sample fit keeps improving as you add order.
- But out-of-sample accuracy stalls, then degrades.
- Coefficients become large, unstable, or barely significant.

The safeguards

- Let AIC/BIC penalize complexity for you.
- Validate on held-out data, not just the training sample.
- When in doubt, choose the simpler model.

Automate, Then Judge

How Box-Jenkins is run today

What the tools do

- Automated search (e.g. `auto_arima`) grids over (p, q) by AIC/BIC.
- It returns a ranked shortlist in seconds.
- This handles the tedious search you once did by hand.

What stays your job

- Making the series stationary correctly (Week 2).
- Reading the residual diagnostics and sanity-checking the pick.
- Judging whether the model makes economic sense.

Box-Jenkins, End to End

Direct Claude Code through the full identify–estimate–diagnose loop

GOAL Run the complete Box-Jenkins loop on a simulated ARMA series, then confirm you recovered the right order.

[INPUT] Prompt · Claude Code

Write and run `box_jenkins.py` that:

1. simulates a known ARMA(1,1) (hide the parameters from the printed output);
2. plots its ACF and PACF and proposes a couple of candidate (p, q) orders;
3. fits each candidate, reports AIC and BIC, and picks the best;
4. runs a Ljung-Box test on the residuals and plots their ACF.

Finally, reveal the true order and tell me whether the loop recovered it.

How to work with it

- Expect ARMA(1,1) to win on AIC/BIC.
- Residual ACF should look like white noise.
- Try an under-fit AR(1) — see the leftover spike.
- Keep it; Part 3 adds forecasting on top.

Deliverable: `box_jenkins.py` + the ACF/PACF and residual plots for one series.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

Both the ACF and PACF tail off gradually. What model class does this suggest?

ARMA(p,q)

Q2

AIC picks ARMA(2,2); BIC picks ARMA(1,1). Which is likely simpler, and why?

BIC — heavier penalty

Q3

Your fitted model's residual ACF still has a spike at lag 1. Next step?

Revise p or q

All three come straight from the cheat sheet, the criteria, and the diagnostic step.

Three Pitfalls to Avoid

Where identification goes wrong

Common mistakes

- "Identify once and trust it."
- "The lowest AIC is always the right model."
- "Skip stationarity and go straight to ARMA."

The reality

- Box-Jenkins is a loop — diagnostics can send you back to step 1.
- AIC can overfit; cross-check with BIC and residual diagnostics.
- ARMA assumes stationarity — difference or detrend first (Week 2).

Three Things to Carry Forward

Part 2 in one screen

1

ARMA fuses feedback and echoes — parsimoniously

A small ARMA(p,q) often matches what a large pure AR or MA would need, capturing persistence and shocks in one compact model.

2

The ACF/PACF cheat sheet identifies the order

A cut-off names the order (ACF→MA, PACF→AR); both tailing off means ARMA. On real data, propose a few candidates and compare.

3

Box-Jenkins is a disciplined loop

Identify, estimate, diagnose — guided by AIC/BIC and parsimony — and repeat until the residuals are white noise.

Next — Part 3: diagnostics in depth, forecasting, and a humbling truth about returns.

★ WEEK 3 · PART 2 · PRACTICE

ARMA Identification, Worked by Hand

Read the ACF and PACF, Then Pass the Diagnostic Gate

The identification cheat sheet, and the Ljung-Box test

Part 2's method, with the arithmetic shown. We apply the ACF-PACF cheat sheet to identify an AR from an MA, computing the partial autocorrelation that cuts off, then run the Box-Jenkins loop and its Ljung-Box diagnostic to confirm the residuals are white noise.

2P

PART 2 · APPLICATIONS

From Plots to a Passing Model

Case A: apply the identification cheat sheet - a PACF that cuts off with a decaying ACF says AR - and compute the partial autocorrelation by hand. Case B: run the Box-Jenkins loop and its Ljung-Box diagnostic to confirm white-noise residuals. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

ARMA combines feedback and echoes, parsimoniously

A small ARMA can replace a large pure AR or MA. The task is choosing the orders p and q .

2

The ACF and PACF identify the orders

An MA cuts off in the ACF; an AR cuts off in the PACF. A decay in both suggests a mixed ARMA.

3

Box-Jenkins is a loop with a diagnostic gate

Identify, estimate, then check the residuals. Only white-noise residuals let you trust the model.

Everything here is a calculator exercise: one PACF cutoff, one Ljung-Box statistic.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the ACF and PACF, or the residuals.
- Read the signature, or run the diagnostic.
- Then decide the model.

What you should watch

- How the PACF cutoff pins down AR.
- How the cheat sheet separates AR, MA, ARMA.
- How Ljung-Box gates the fit.

Identifying the Order

Read the cheat sheet

Given a return series, the ACF and PACF tell you which model generated it. The cheat sheet is simple: an MA cuts off in the ACF, an AR cuts off in the PACF, and a mix decays in both. We take the AR(1) from Part 1, compute its partial autocorrelation, and watch it cut off at lag one - confirming an AR while its ACF still decays

The candidate

- An AR(1) with coefficient 0.6.
- Its ACF decays: 0.6, 0.36, 0.22.
- What does its PACF do?

What we'll do by hand

- Compute the partial autocorrelation.
- See it cut off after lag one.
- Read the cheat sheet to identify AR.

ARMA and Its Signatures

Feedback and echoes in one line

$$\phi(L) y_t = \theta(L) \varepsilon_t \quad (\text{ARMA: AR feedback and MA shock-echoes in one compact line})$$

The two polynomials

- The AR polynomial captures feedback.
- The MA polynomial captures shock echoes.
- Together, a parsimonious ARMA.

The identification idea

- Each part leaves a fingerprint.
- AR shows in the PACF, MA in the ACF.
- Read the fingerprints, recover the orders.

The PACF Cuts Off

The AR fingerprint

$$\phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} = \frac{0.36 - 0.36}{1 - 0.36} = 0 \Rightarrow \text{PACF cuts off after lag 1}$$

The calculation

- The lag-1 partial is just 0.6.
- The lag-2 partial works out to zero.
- And every lag beyond stays zero.

What it shows

- The PACF cuts off sharply after lag one.
- Meanwhile the ACF only decays.
- That pairing is the AR(1) signature.

The Identification Cheat Sheet

AR, MA, or ARMA

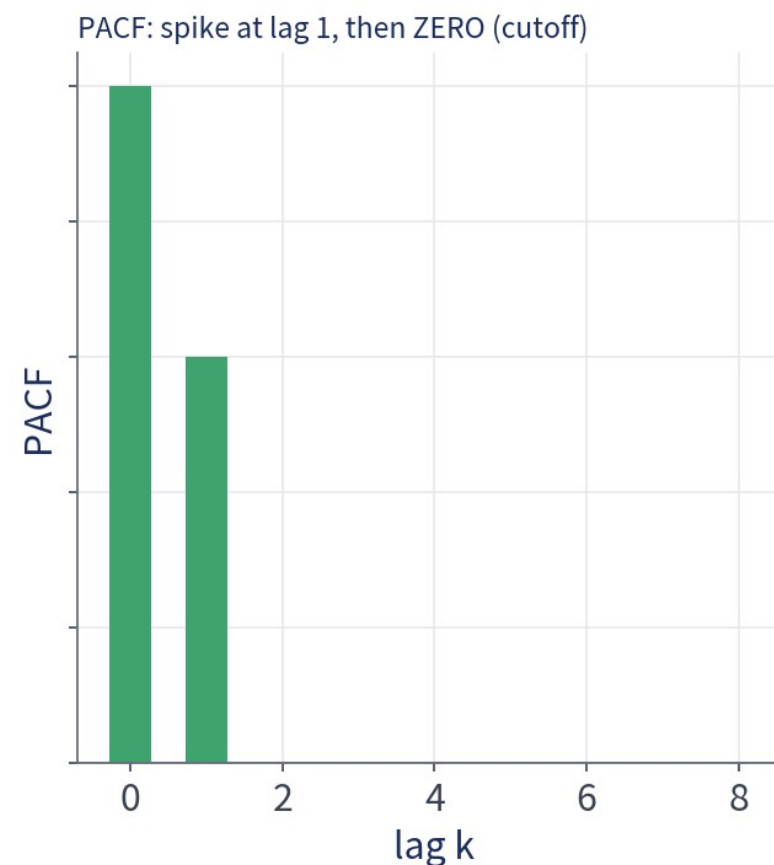
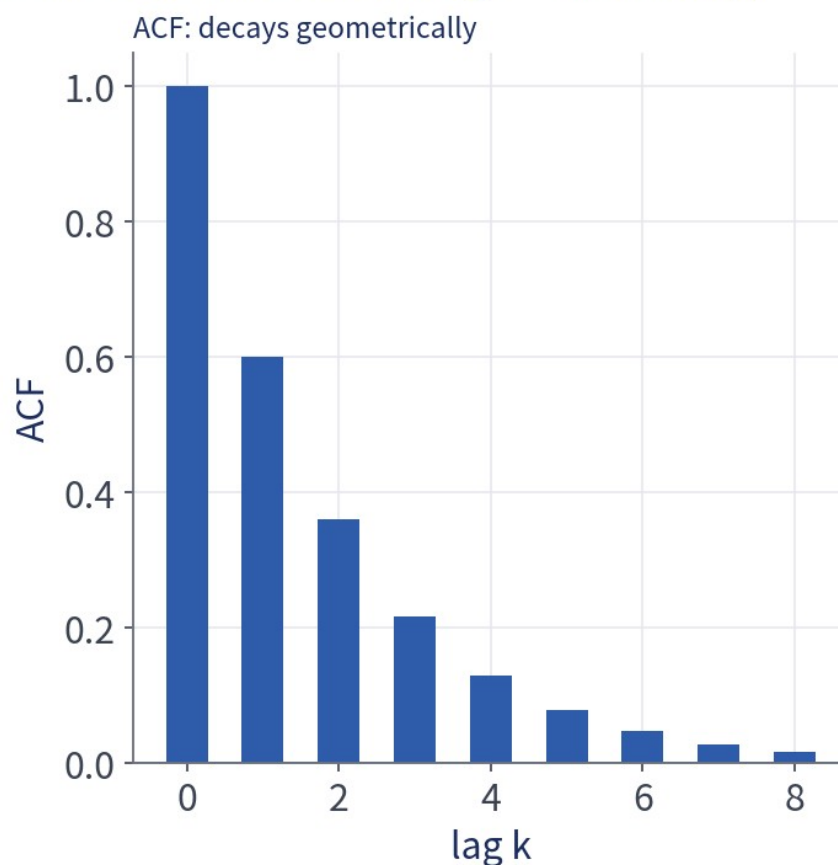
model	ACF	PACF	verdict here
AR(p)	decays	cuts off at p	our case: AR(1)
MA(q)	cuts off at q	decays	the mirror image
ARMA(p,q)	decays	decays	both tail off

Our series decays in the ACF and cuts off at lag one in the PACF - the top row - so it's an AR(1). An MA(1) would be the mirror: a sharp ACF cutoff with a decaying PACF. When both merely decay, suspect a mixed ARMA.

The AR Fingerprint

Left, the ACF decays geometrically - never quite zero. Right, the PACF spikes once at lag one and then drops to exactly zero. A decaying ACF paired with a PACF that cuts off is the unmistakable signature of an AR(1): the cheat sheet made visible

Case A: ACF decays + PACF cuts off at lag 1 $\Rightarrow$ this is an AR(1)



Why Parsimony Wins

The simplest model that fits

The parsimony principle

- Prefer the fewest parameters that work.
- A tight ARMA beats a bloated pure model.
- Fewer parameters forecast more reliably.

Reading mixed cases

- If both ACF and PACF decay, it's ARMA.
- Start small and add terms only if needed.
- The plots suggest; the diagnostics decide.

Box-Jenkins and the Gate

Estimate, check, and delete

Identification is only the first step of the Box-Jenkins loop. After estimating the coefficients, you must check the residuals: if the model captured all the structure, what's left should be white noise. The Ljung-Box test turns the residual autocorrelations into a single statistic. We compute it and see the model pass - and see what

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The loop

- Identify from the plots.
- Estimate by maximum likelihood.
- Diagnose the residuals - then repeat if needed.

What we'll do by hand

- Take the residual autocorrelations.
- Compute the Ljung-Box statistic.
- Compare to the chi-square cutoff.

The Box-Jenkins Loop

A disciplined cycle, not a recipe

identify (ACF/PACF) → **estimate** (MLE) → **diagnose** (residuals) ↺

The three stages

- Identify orders from ACF and PACF.
- Estimate coefficients by likelihood.
- Diagnose the residuals for structure.

The gate

- If residuals aren't white noise, go back.
- Adjust the orders and re-estimate.
- Only a clean pass earns your trust.

The Ljung-Box Statistic

Are the residuals white noise?

$$Q = T(T + 2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k} \sim \chi_{m-\text{params}}^2 \quad (\text{are the residuals white noise?})$$

What it sums

- The squared residual autocorrelations.
- Weighted and scaled by the sample size.
- Across the first several lags.

The distribution

- Under white noise, it's chi-square.
- Degrees of freedom: lags minus parameters.
- A large Q means leftover structure.

Pass or Fail

The diagnostic verdict

$$Q = 1.29 < \underbrace{9.49}_{\chi^2_4 \text{ 5\%}} \Rightarrow \text{residuals are white noise} \Rightarrow \text{the model passes}$$

Our model passes

- The residual autocorrelations are tiny.
- Q is just 1.29, well under 9.49.
- The residuals are white noise.

A failing model

- Leftover autocorrelation of 0.28, 0.20, ...
- Pushes Q to 32.5, far past the cutoff.
- Reject it - the orders were wrong.

Residuals That Behave

Every residual autocorrelation sits comfortably inside the significance bands - no lag pokes through. That's what white noise looks like: the model has extracted all the linear structure, leaving only unpredictable shocks. This is the diagnostic gate the Ljung-Box statistic formalizes

Case B: residual ACF sits inside the bands - white noise, the model passes



The Gate You Can't Skip

Diagnostics before trust

Why diagnose

- A good in-sample fit can still miss structure.
- Leftover autocorrelation means a wrong order.
- Ljung-Box catches it in one number.

Parsimony and selection

- Among passing models, prefer the simplest.
- AIC and BIC break ties by penalizing parameters.
- The tightest adequate model wins.

The Identification Method

Plots in, a trusted model but

The routine is Box-Jenkins. Read the ACF and PACF to propose orders - a PACF cutoff for AR, an ACF cutoff for MA, both decaying for ARMA. Estimate the coefficients. Then diagnose: the Ljung-Box test must confirm white-noise residuals before you trust anything. Among models that pass, the simplest wins. Plots suggest,

The identification (Case A)

- ACF decays, PACF cuts off at one.
- So the model is AR(1).
- The cheat sheet in action.

ic

The diagnostic (Case B)

- Ljung-Box Q of 1.29 beats 9.49.
- Residuals are white noise.
- The model passes the gate.

Reproduce the Identification and Diagnostic

Direct Claude Code to check your arithmetic

GOAL Identify an AR from ACF/PACF, then run the Ljung-Box diagnostic on residuals.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - identify:

- * simulate an AR(1) with $\phi=0.6$;
- * plot the ACF and PACF and confirm the PACF cuts off at lag 1 while the ACF decays.

CASE B - diagnose:

- * for residual ACF $[0.05, -0.03, 0.04, -0.02, 0.03]$, $T=200$, compute the Ljung-Box Q and compare to the chi-square cutoff ($df = 5 - 1$);
- * repeat with $[0.28, 0.20, 0.15, 0.11, 0.08]$ and reject.

Check against the slides

- PACF $\phi_{22} = 0$.
- ACF decays, PACF cuts off.
- Ljung-Box $Q = 1.29 < 9.49$.
- bad model $Q = 32.5 > 9.49$.

Deliverable: `part2_practice.py` whose printed numbers match the PACF cutoff and the Ljung-Box verdicts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you spot an AR from the plots?

A PACF that cuts off with a decaying ACF

Q2 What are the three Box-Jenkins stages?

Identify, estimate, and diagnose - in a loop

Q3 What does the Ljung-Box test check?

Whether the residuals are white noise

All three come from the cheat sheet, the Box-Jenkins loop, and the Ljung-Box diagnostic.

Three Pitfalls to Avoid

Where identification trips people

Common mistakes

- "A great fit means the model is adequate."
- "Add AR and MA terms until it fits perfectly."
- "Skip the residual check and forecast."

The reality

- Only white-noise residuals prove adequacy.
- Over-fitting hurts forecasts - stay parsimonious.
- The Ljung-Box gate is not optional.

Two Calculations to Remember

Part 2 applications in one screen

1

The PACF cutoff identifies the AR

The AR(1)'s partial autocorrelation was 0.6 at lag one and exactly zero after - a sharp cutoff - while its ACF only decayed. That pairing, read from the cheat sheet, identifies an AR(1): a PACF cutoff plus a decaying ACF.

2

Ljung-Box gates the model

The residual autocorrelations were tiny, giving a Ljung-Box Q of 1.29, well under the chi-square cutoff of 9.49 - the residuals are white noise, so the model passes. A model with leftover structure gave Q of 32.5 and was rejected.

3

Plots suggest, diagnostics decide

Box-Jenkins is a loop: identify orders from the ACF and PACF, estimate, then diagnose the residuals. Among models that pass the gate, the most parsimonious - by AIC or BIC - is the one to trust.

Next — Part 3 practice: full residual diagnostics and forecasting from a fitted ARMA model.

★ WEEK 3 · PART 3 of 3

Selection, Diagnostics & Forecasting

Validating a Model — and Knowing Its Limits

AIC/BIC · residual diagnostics · forecasting · the humbling truth about returns

A fitted model is not a finished model. Part 3 completes the loop: choose among candidates, prove the residuals are clean, forecast honestly — and confront why ARMA finds so little in raw returns.

3

PART 3 · VALIDATION

From a Fit to a Forecast

Model selection with AIC and BIC, diagnostic checking of the residuals, forecasting with honest intervals, and a clear-eyed look at what ARMA can and cannot predict.

Closing the Box-Jenkins Loop

Part 2 fit models; Part 3 validates and uses them

1

We can identify and estimate an ARMA

The ACF/PACF propose a candidate; the software fits it. But a fit is only a hypothesis until it survives scrutiny.

2

Now we select, check, and forecast

Compare candidates with AIC/BIC, confirm the residuals are white noise, and only then produce forecasts with honest uncertainty.

3

And we stay humble

ARMA is powerful where linear memory exists — and nearly useless on raw returns. Knowing the difference is the mark of an expert.

Part 3 in one line: prove the model is clean, forecast honestly, and know its limits.

AIC and BIC, Revisited

Two penalties, two philosophies

Both criteria reward fit and punish complexity — they differ only in how harshly. AIC aims for the best forecaster; BIC aims for the true, simplest model.

AIC — the forecaster's choice

- A lighter penalty ($2k$) tolerates a few extra terms.
- It targets out-of-sample predictive accuracy.
- It can occasionally keep a parameter that only fits noise.

BIC — the truth-seeker's choice

- A heavier penalty ($k \cdot \ln T$) that grows with sample size.
- It is consistent: it finds the true order as data grows.
- It leans toward simpler, more parsimonious models.

AIC vs BIC: Which and When

A quick decision guide

Criterion	Penalty on k	Tends to select	Prefer it when
AIC	$2k$ (lighter)	slightly larger models	your goal is forecasting
BIC	$k \cdot \ln T$ (heavier)	simpler models	your goal is the 'true' order

When they disagree, default to the simpler (BIC) model unless forecasting clearly needs the extra terms.

Are the Residuals White Noise?

The final gate before trusting a model

A model has captured the structure only if what it leaves behind — the residuals — is indistinguishable from white noise. Any leftover pattern means the job is not done.

What 'clean' means

- No autocorrelation left in the residuals at any lag.
- Roughly constant variance and no obvious outliers.
- Nothing the model could still have exploited.

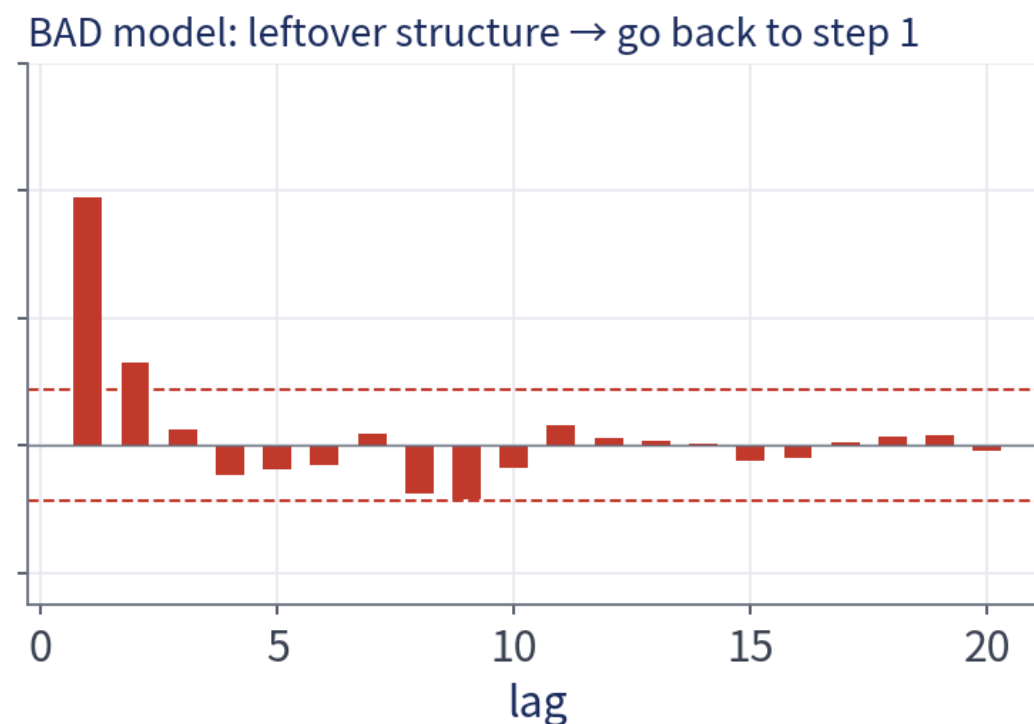
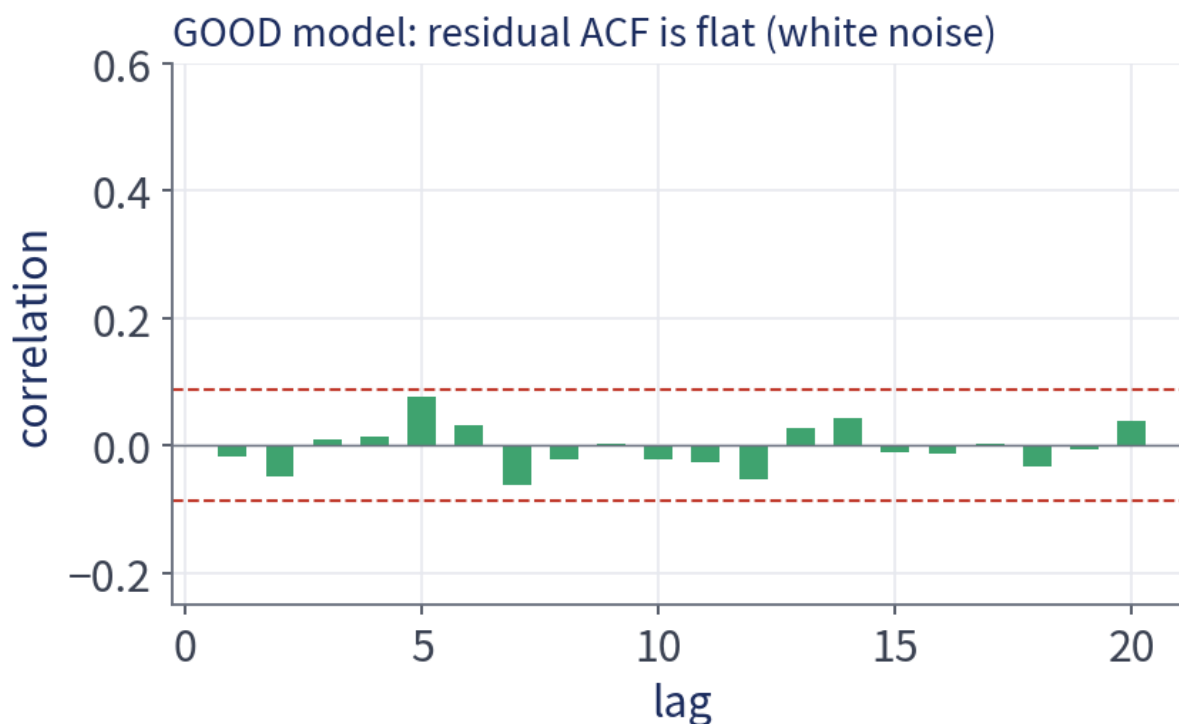
Why it is the exit test

- If structure remains, the forecast is leaving signal on the table.
- A leftover spike tells you exactly where to add order.
- Only white-noise residuals let you leave the loop.

Reading the Residual ACF

A good model leaves a flat residual ACF (left). A leftover spike (right) means the model missed structure — go back and raise p or q .

The final gate: are the residuals indistinguishable from white noise?



The Ljung-Box Test

A formal check for leftover autocorrelation

$$Q = T(T + 2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{T-k} \sim \chi_{h-m}^2 \quad (\text{Ljung-Box})$$

What it does

- Combines the first h residual autocorrelations into one statistic.
- Under a clean model it follows a chi-squared distribution.
- A small p-value means the residuals still carry structure.

How to use it

- Run it on the residuals after fitting.
- A LARGE p-value is good — it means 'looks like white noise'.
- Pair it with the residual ACF plot; never rely on one alone.

Other Diagnostic Checks

White noise is necessary, not sufficient

Also inspect

- Normality of residuals: a Q-Q plot or a Jarque-Bera test.
- Constant variance: watch for volatility clustering in residuals.
- Stability: do the coefficients hold across sub-samples?

What they reveal

- Fat-tailed residuals warn that intervals may be too narrow.
- Clustered residual variance points to a GARCH model (later).
- Unstable coefficients hint at a structural break.

Forecasting With ARMA

Turn the fitted model forward in time

To forecast, feed the model's own predictions back into it, step by step. Each step uses known values and past forecasts — and expected future shocks are simply zero.

The mechanism

- One step ahead uses the last observed values directly.
- Further steps substitute earlier forecasts for unknown values.
- Unknown future shocks contribute their expected value: zero.

The consequence

- The influence of what you know fades with the horizon.
- So long-horizon forecasts converge to the series' mean.
- The next slides make this concrete.

Forecasts Decay to the Mean

The farther out you look, the less the past helps

$$\hat{X}_{T+h} = \mu + \phi^h (X_T - \mu) \xrightarrow{h \rightarrow \infty} \mu \quad (\text{AR}(1) \text{ forecast})$$

What the formula says

- The forecast is the mean plus a shrinking piece of today's deviation.
- That piece shrinks like ϕ^h as the horizon h grows.
- Beyond a few steps, the best guess is essentially the mean.

Why it makes sense

- A stationary series has no lasting trend to extrapolate.
- Its memory of any shock fades — so must the forecast's edge.
- This is mean reversion, seen through the forecast.

Forecasts Need Honest Intervals

A point forecast alone is dangerously incomplete

$$\hat{X}_{T+h} \pm z_{0.975} \sigma_h, \quad \sigma_h \text{ grows with the horizon } h$$

Why intervals matter

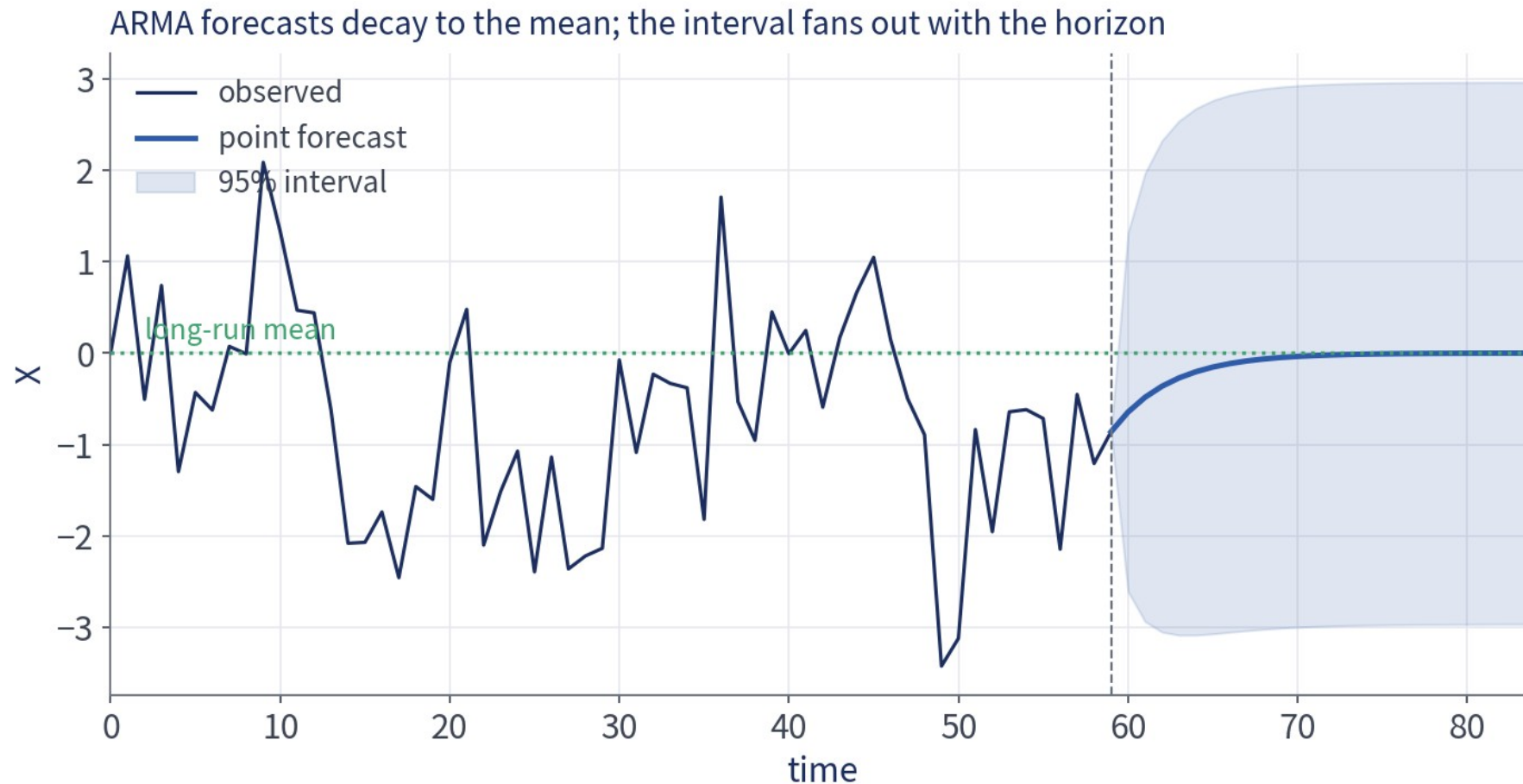
- A single number hides how uncertain the forecast really is.
- The interval σ_h widens as the horizon grows.
- It reflects the accumulation of future unknown shocks.

How to read them

- Near term: tight bands — recent data pins the value down.
- Far term: wide bands centered on the mean.
- Report the interval, not just the point, every time.

The Forecast Fan

The point forecast (blue) decays to the long-run mean while the 95% interval fans out — a compact picture of everything ARMA forecasting does.



How Far Ahead Is Useful?

Knowing when a forecast stops saying anything

An ARMA forecast is informative only while its point estimate still differs meaningfully from the mean. Past that horizon, you are just quoting the average.

The practical limit

- Persistence sets the useful horizon: higher ϕ reaches farther.
- Once ϕ^h is tiny, the forecast is indistinguishable from the mean.
- The interval, meanwhile, has widened to its long-run width.

The honest takeaway

- Short horizons: genuine, usable information.
- Long horizons: the mean, dressed up with wide bands.
- Match your forecast horizon to the series' memory.

In-Sample vs Out-of-Sample

The only test that really counts

A model can fit the past beautifully and still forecast the future badly. The verdict that matters is out-of-sample: performance on data the model never saw while fitting.

The trap of in-sample fit

- In-sample error always falls as you add parameters.
- That improvement can be pure overfitting.
- It flatters complex models that will not generalize.

The out-of-sample discipline

- Hold out the most recent data; fit on the rest.
- Forecast the held-out period and measure the error.
- Or roll the window forward for a rigorous back-test.

Forecast Error Metrics

How to score a forecast

Metric	What it measures	Character
RMSE	root mean squared error	penalizes large misses heavily
MAE	mean absolute error	average miss; robust to outliers
MAPE	mean absolute percentage error	scale-free, but unstable near zero

Always score against a naive benchmark (e.g. 'tomorrow = today'), never against zero error.

The Benchmark You Must Beat

A forecast is only good if it beats the obvious

For many financial series the naive forecast — 'tomorrow equals today' — is remarkably hard to beat. If your ARMA cannot beat it out of sample, it adds nothing.

The naive benchmarks

- Random walk: next value equals the last observed value.
- Historical mean: next value equals the long-run average.
- Seasonal naive: next value equals the same period last cycle.

The honest comparison

- Report your model's error AND the benchmark's error.
- Beating the benchmark is the real evidence of value.
- Failing to beat it is a useful, humbling result.

A Humbling Truth About Returns

ARMA describes structure — but returns have little

Apply the entire Box-Jenkins machine to raw asset returns and you find almost nothing: their autocorrelation is essentially zero. The direction of returns is not linearly predictable.

Why this happens

- Efficient markets arbitrage away simple linear patterns (Week 1).
- So the ACF of returns sits inside the confidence bands.
- An ARMA on returns fits essentially white noise.

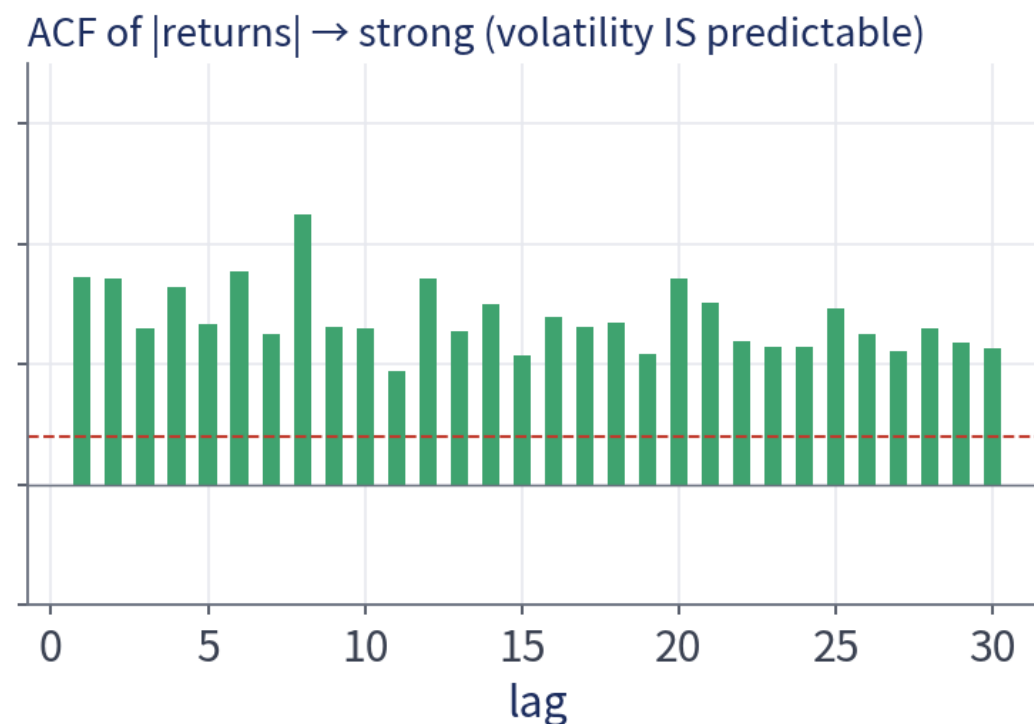
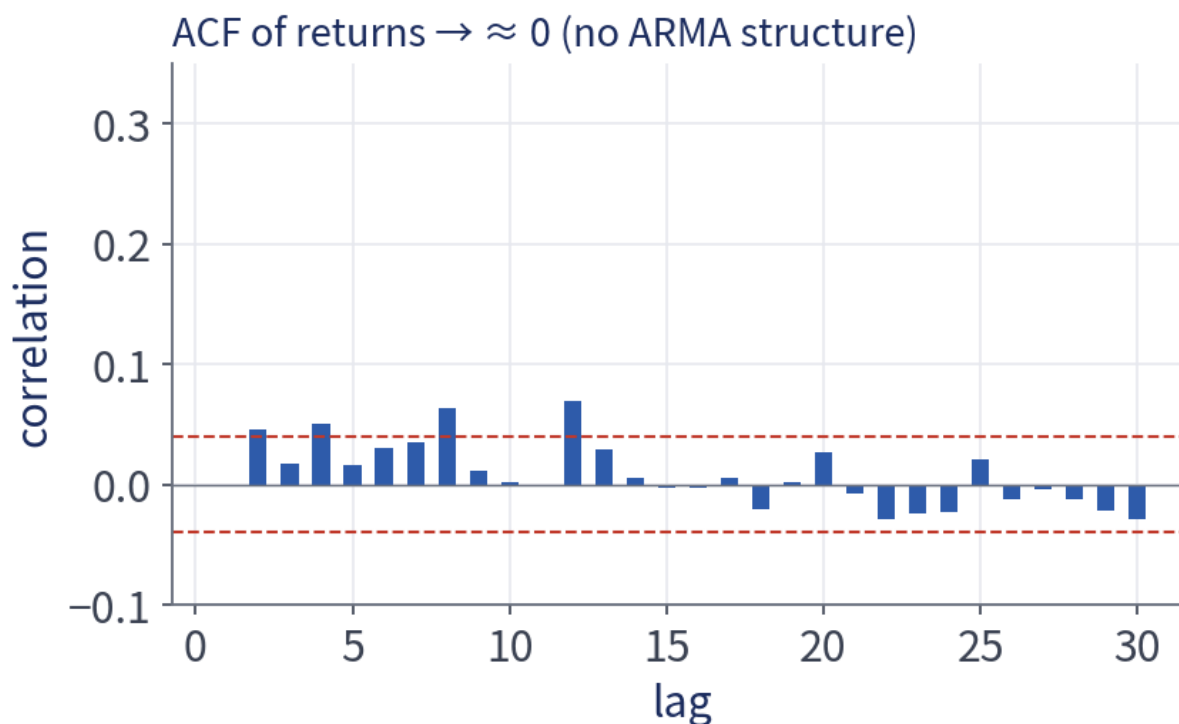
Why it is not defeat

- It tells you where NOT to spend effort — direction forecasting.
- It redirects you to what IS predictable.
- That redirection is the most valuable lesson of the week.

Returns Resist ARMA — But Volatility Does Not

Left: the ACF of returns is essentially zero, so ARMA finds nothing. Right: the ACF of $|\text{returns}|$ is strongly positive — the predictable structure lives in volatility.

The humbling truth: ARMA finds little in returns — the structure lives in volatility



So What IS Predictable? Volatility

The door ARMA opens onto later weeks

$$\underbrace{\rho_r(k) \approx 0}_{\text{returns: no ARMA}} \quad \underbrace{\rho_{|r|}(k) > 0}_{\text{volatility: predictable}}$$

The split, one more time

- The level of returns: no linear memory — ARMA is powerless.
- The magnitude of returns: strong, persistent memory.
- Volatility clusters, and clustering is exactly what we can model.

Where it leads

- Model the variance, not the mean: ARCH and GARCH.
- The same ACF logic, applied to squared or absolute returns.
- Week 3's tools carry straight over to volatility.

Where ARMA Helps — and Where It Does Not

ARMA earns its keep on series with genuine linear memory; it finds little in raw returns, but points the way to volatility models.

Where ARMA Helps - and Where It Does Not

A model is only as good as the structure that is actually there to find.

ARMA earns its keep here

Series with genuine linear memory:

- macro levels & growth rates
- interest rates and spreads
- inflation, demand, inventories
- cointegrated spreads (Week 2)
- seasonal / trending data (Week 4)

These have autocorrelation the ACF / PACF can see - so ARMA can exploit it.

The humbling truth: raw returns

Asset returns are almost uncorrelated:

- ACF of returns is essentially 0
- so ARMA finds little to model
- direction is not linearly predictable

But not all is lost:

The ACF of $|returns|$ is strongly positive - volatility IS predictable.

That is the door to GARCH (later weeks).

The Complete Modeling Checklist

Weeks 2 and 3 in one routine

1 · Stationarize

Week 2

Difference or detrend;
confirm with ADF/KPSS.

2 · Identify

ACF/PACF

Propose candidate (p, q)
from the fingerprints.

3 · Estimate

fit + rank

Fit candidates;
compare by AIC and BIC.

4 · Diagnose

residuals

Check the residual ACF
and Ljung-Box.

5 · Forecast

honestly

Predict with intervals;
beat a naive benchmark.

One disciplined pass from raw data to a validated, honestly-evaluated forecast.

Forecast, Evaluate, and Confront Returns

Direct Claude Code from a fitted model to an honest verdict

GOAL Forecast a fitted ARMA with intervals, back-test it against a naive benchmark, then test the humbling truth on real returns.

[INPUT] Prompt · Claude Code

Write and run `forecast_lab.py` that:

1. fits your chosen ARMA (from Lab 2) and produces an $h=20$ forecast with 95% bands;
2. back-tests it out-of-sample vs a random-walk benchmark, reporting RMSE for both;
3. then fits an ARMA to real daily RETURNS and plots the ACF of returns vs $|returns|$.

Tell me: does the ARMA beat the naive benchmark, and does it find any structure in the returns themselves?

How to work with it

- The forecast should fan out and revert to the mean.
- Beating the random walk is hard — note if it fails.
- The returns' ACF should look flat; $|returns|$ should not.
- This closes the Week-3 arc and opens volatility.

Deliverable: `forecast_lab.py` + a forecast fan and the returns-vs- $|returns|$ ACF.

Three Questions Before You Move On

If you can answer these, Part 3 has landed

Q1

A Ljung-Box test on your residuals gives $p = 0.02$. Good or bad news?

Bad — structure left

Q2

You forecast an AR(1) 30 steps ahead. What does the point forecast approach?

The mean

Q3

Your ARMA on returns has a near-flat ACF. What does that tell you?

No linear structure

All three follow from diagnostics, mean reversion, and the humbling truth.

Three Pitfalls to Avoid

Where forecasting goes wrong

Common mistakes

- "A great in-sample fit means great forecasts."
- "Report the point forecast; the interval is optional."
- "ARMA should be able to predict returns."

The reality

- Only out-of-sample performance counts; in-sample flatters complexity.
- The interval is the forecast — a point alone hides the uncertainty.
- Raw returns are ~unpredictable; model volatility instead.

Week 3 in Three Ideas

Pulling the whole week together

1

AR and MA are two engines; ARMA fuses them

Part 1: AR remembers levels, MA remembers shocks, and Wold's theorem plus $AR \leftrightarrow MA$ duality tie them together. The ACF/PACF are the fingerprints.

2

Box-Jenkins turns modeling into a disciplined loop

Part 2: identify with the cheat sheet, estimate, select by AIC/BIC, and diagnose — repeating until the residuals are white noise.

3

Forecast honestly, and know ARMA's limits

Part 3: forecasts revert to the mean with widening intervals; ARMA finds little in raw returns, pointing us toward volatility models next.

Next — Week 4: ARIMA and Seasonality, extending ARMA to trending and seasonal data.

★ WEEK 3 · PART 3 · PRACTICE

ARMA Forecasting, Worked by Hand

Watch the Forecast Fade to the Mean, and the Band Fan Out

Point forecasts, widening intervals, and the benchmark

Part 3's payoff, with the arithmetic shown. We turn a fitted AR(1) forward, watch its point forecast decay geometrically to the mean, see the prediction interval widen to the unconditional level - and confront the humbling truth about forecasting returns.

3P

PART 3 · APPLICATIONS

How Far Ahead Is Useful

Case A: iterate the fitted AR(1) forward and watch the point forecast decay to the mean as the past stops helping.
Case B: the forecast error variance grows, widening the band to the unconditional level - and a forecast is only good if it beats the naive benchmark. Then a Claude Code lab.

What Part 3 Told Us

Now we apply it - with numbers

1

Forecasts decay to the mean

A stationary model's forecast reverts to its long-run mean. The farther out you look, the less the last value helps.

2

Intervals must widen honestly

Uncertainty grows with the horizon toward the unconditional variance. A point forecast without a band is dangerously incomplete.

3

You must beat the naive benchmark

A forecast only counts if it beats the obvious one. For returns, with little structure, that bar is humbly hard to clear.

Everything here is a calculator exercise: one decaying forecast, one widening band, one benchmark.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a fitted AR(1) and its last value.
- Iterate the forecast, or grow the variance.
- Then read the horizon.

What you should watch

- How the forecast decays to the mean.
- How the band widens to the unconditional level.
- How hard it is to beat naive on returns.

The Forecast Decays

The past stops helping

With a fitted AR(1), forecasting is just iterating the model forward. The last value starts above the mean, but each step multiplies its deviation by the coefficient, so the forecast decays geometrically back to the long-run mean. We compute the point forecast at several horizons and watch the memory fade

The fitted model

- Coefficient 0.6, mean 1.0.
- The last observed value is 3.0.
- Two full points above the mean.

What we'll do by hand

- Iterate the forecast forward.
- Watch the deviation shrink by 0.6 each step.
- See it decay toward the mean.

The ARMA Forecast

Turn the fitted model forward

$$\hat{y}_{T+h} = \mu + \phi^h(y_T - \mu) \quad (\text{iterate the fitted model forward; the deviation shrinks by } \phi \text{ each step})$$

How it iterates

- The forecast is the mean plus a decayed deviation.
- Each step multiplies the gap by the coefficient.
- Future shocks are unknown, so set to zero.

The consequence

- The deviation from the mean shrinks geometrically.
- Distant forecasts carry little of the last value.
- The model forgets, at rate ϕ .

The Point Forecasts

Fading toward the mean

$$1.0 + 2(0.6)^h : \quad \mathbf{2.20} \ (h=1), \ \mathbf{1.72} \ (h=2), \ \mathbf{1.43} \ (h=3) \rightarrow \mu=1.0$$

The arithmetic

- One step: 1.0 plus 2 times 0.6, or 2.2.
- Two steps: 1.0 plus 2 times 0.36, or 1.72.
- Three steps: 1.43, and falling.

The trajectory

- Each forecast is closer to the mean.
- The initial gap of 2.0 keeps shrinking.
- By ten steps it's essentially at 1.0.

Decay to the Mean

When the past stops mattering

$$\text{as } h \rightarrow \infty : \phi^h \rightarrow 0 \Rightarrow \hat{y}_{T+h} \rightarrow \mu \quad (\text{the past stops helping})$$

The limit

- As the horizon grows, ϕ -to-the- h vanishes.
- So the forecast converges to the mean.
- The last value's influence disappears.

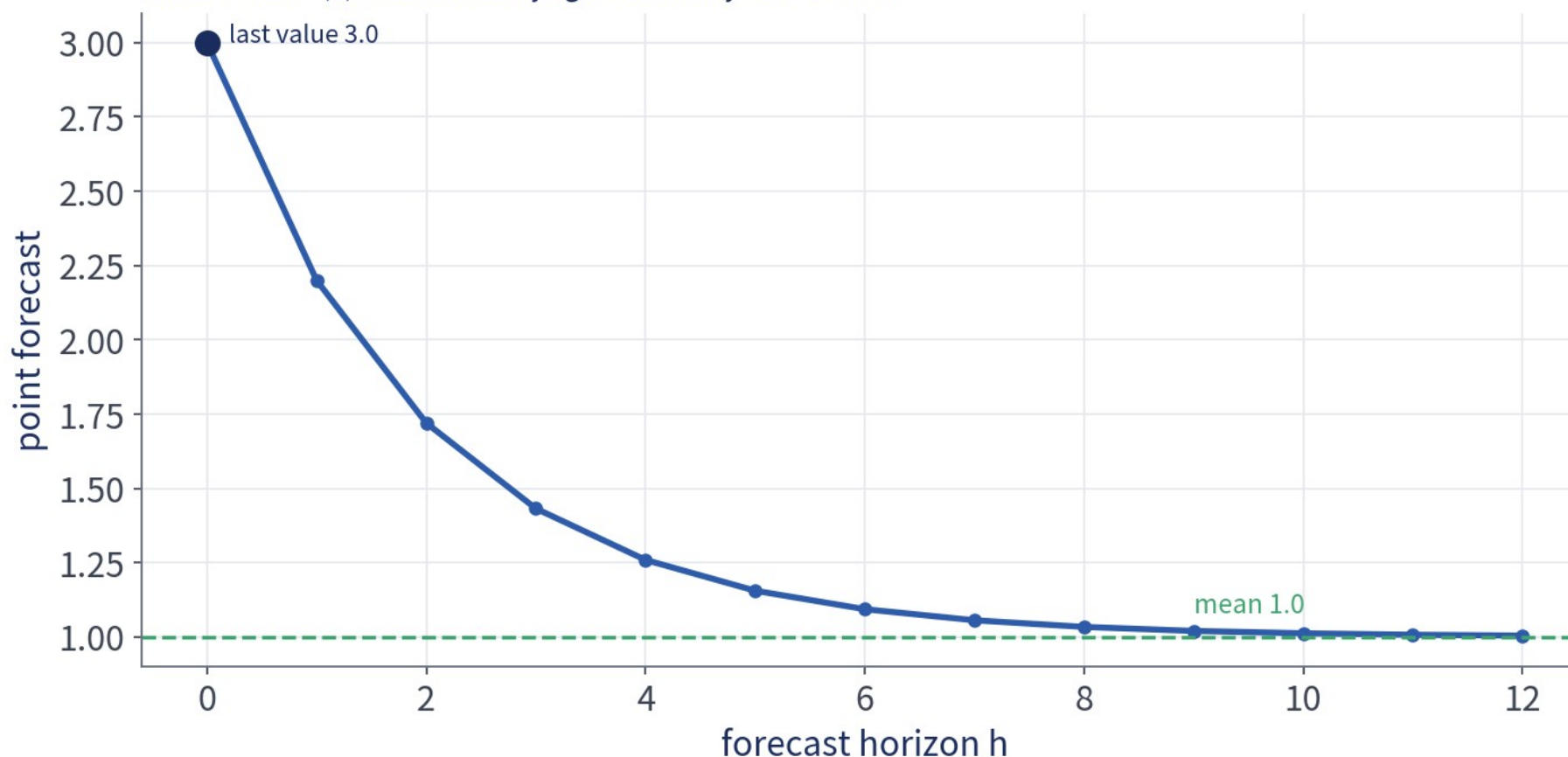
The lesson

- There's a horizon past which you learn nothing.
- Beyond it, the best guess is just the mean.
- Know where your forecast stops speaking.

The Memory Fades

Starting from a last value of 3.0, two points above the mean, the forecast decays smoothly back toward the dashed mean of 1.0. Each step erases 40% of the remaining gap. Within about ten steps the forecast is indistinguishable from the mean – the point where the past has nothing left to say

Case A: the AR(1) forecast decays geometrically to the mean



The Information Horizon

How far the model can see

What decay implies

- Short horizons carry real information.
- Long horizons collapse to the mean.
- The coefficient sets how fast.

How to use it

- Trade on the near-term forecast, not the far one.
- A high coefficient means a longer useful horizon.
- Past that, the mean is your best guess.

Honest Intervals, Hard Benchmark

The band around the bar grows wider

A point forecast alone is dangerously incomplete. As the horizon grows, forecast errors accumulate, so the prediction interval widens - toward the unconditional variance. We compute that widening band, then confront the benchmark: a forecast is only useful if it beats the naive one, a bar that returns, with their thin

ra

The widening band

- Error variance grows with the horizon.
- It approaches the unconditional variance.
- So the band fans out, then levels off.

What we'll do by hand

- Compute the error variance at each horizon.
- Turn it into a widening interval.
- Then weigh the benchmark.

The Forecast Error Variance

Uncertainty that accumulates

$$\text{Var}(e_h) = \sigma^2 \frac{1-\phi^{2h}}{1-\phi^2} \nearrow \frac{\sigma^2}{1-\phi^2} = 1.5625 \quad (\text{the unconditional variance})$$

How it grows

- Each step adds another shock's variance.
- The sum is a geometric series.
- It climbs toward a finite limit.

The ceiling

- The limit is the unconditional variance.
- Here, 1.5625.
- Beyond it, uncertainty can't grow.

The Widening Band

From one step to the limit

horizon	error variance	std dev	95% band
$h = 1$	1.00	1.00	+/- 1.96
$h = 2$	1.36	1.17	+/- 2.29
$h = 3$	1.49	1.22	+/- 2.39
$h = \infty$	1.5625	1.25	+/- 2.45

The error variance climbs from 1.0 at one step toward the unconditional 1.5625, so the 95% band widens from plus-or-minus 1.96 to plus-or-minus 2.45. Far out, the forecast is just the mean, plus or minus the unconditional spread.

The Benchmark, and Returns

The bar, and the humbling truth

95% band : ± 1.96 ($h=1$) $\rightarrow \pm 2.45$ ($h=\infty$); must beat the naive/mean forecast

The benchmark rule

- Compare to the naive or mean forecast.
- Beating it is the minimum standard.
- In-sample fit doesn't count.

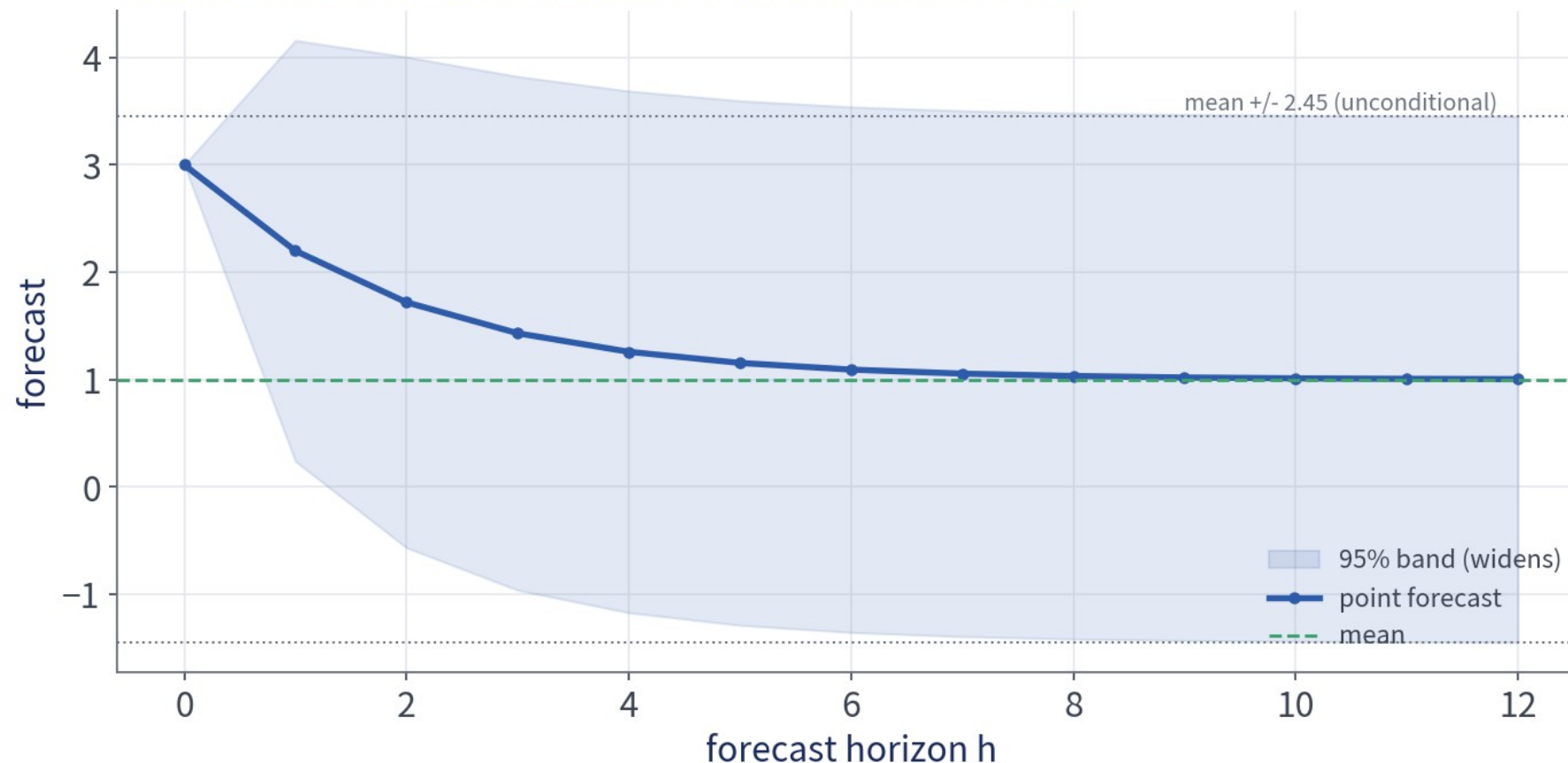
The humbling truth

- Returns carry very little ARMA structure.
- So the mean forecast is hard to beat.
- Honesty about this saves real money.

The Band Fans Out

The point forecast (blue) decays to the mean while its 95% band widens with the horizon, approaching the dotted unconditional limits of the mean plus or minus 2.45. Far out, the forecast says only 'the mean, give or take the unconditional spread' - which is to say, almost nothing

Case B: bands widen to the unconditional sd - far out, only the mean remains



Forecasting Returns, Honestly

Structure is real, but thin

Where ARMA earns its keep

- Volume, volatility, and macro series have structure.
- There, the forecast beats naive.
- The band tells you how much to trust it.

Where it humbles you

- Return levels are close to a random walk.
- The mean forecast is brutally hard to beat.
- Admitting this is the mark of a pro.

The Forecasting Checklist

Point, interval, benchmark, horizon

Forecasting from a fitted ARMA has a fixed discipline. Iterate the point forecast forward, knowing it decays to the mean. Attach an honest interval that widens to the unconditional spread. Judge it out of sample against the naive benchmark - and never trust the in-sample fit alone. And ask how far ahead the forecast still says

The forecast (Case A)

- Decays 2.2, 1.72, 1.43 to the mean 1.0.
- The last value's influence fades.
- A finite information horizon.

The interval (Case B)

- Band widens from 1.96 to 2.45.
- Levels off at the unconditional spread.
- Must still beat naive to matter.

Reproduce the Forecast and the Band

Direct Claude Code to check your arithmetic

GOAL Iterate the AR(1) forecast, build the widening interval, and beat a benchmark.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that, for an AR(1)
with phi=0.6, mu=1.0, sigma^2=1, last value 3.0:
CASE A - point forecast:
    * print y_hat for h = 1..10 and confirm it decays
      to the mean.
CASE B - interval and benchmark:
    * print the error variance and 95% band for
      h = 1, 2, 3, and the limit;
    * compare an ARMA forecast to the mean forecast
      on a simulated return series (expect little gain).
Then plot the decaying forecast with its widening band.
```

Check against the slides

- forecast 2.2, 1.72, 1.43 -> 1.0.
- band +/-1.96 -> +/-2.45.
- variance limit 1.5625.
- returns barely beat the mean.

Deliverable: part3_practice.py whose printed numbers match the forecast decay and the widening band.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Where does a stationary ARMA forecast head?

Toward the long-run mean, geometrically

Q2 What does the prediction band approach?

The unconditional standard deviation

Q3 What must a forecast beat to be useful?

The naive or mean benchmark, out of sample

All three come from the forecast decay, the widening interval, and the benchmark rule.

Three Pitfalls to Avoid

Where forecasting trips people

Common mistakes

- "A long-horizon forecast is still informative."
- "Report the point forecast without a band."
- "A good in-sample fit means good forecasts."

The reality

- Far out, the forecast is just the mean.
- Always attach a widening interval.
- Only an out-of-sample win over naive counts.

Two Calculations to Remember

Part 3 applications in one screen

1

The forecast decays to the mean

From a last value of 3.0, the AR(1) forecast fell to 2.2, then 1.72, then 1.43 - each step erasing 40% of the gap - converging to the mean of 1.0. Past a horizon, the last value tells you nothing and the mean is the best guess.

2

The band widens to the unconditional spread

The error variance climbed from 1.0 toward the unconditional 1.5625, so the 95% band widened from plus-or-minus 1.96 to plus-or-minus 2.45. Far out, the forecast is only the mean plus or minus that unconditional spread.

3

Beat the benchmark, or stay humble

A forecast counts only if it beats naive out of sample - and returns, with almost no ARMA structure, rarely do. ARMA shines on volume and volatility; on return levels, honesty beats false confidence.

Week 3 complete — from AR and MA through identification to forecasting, its intervals, and its honest limits.